

The Unification of Formation, Motion, Decay, and Thermalization of Matter from the Bost–Connes System

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Abstract

The formation, motion, decay, and thermalization of matter have each been established on the state space of the Bost–Connes system, but the complete cycle map has not yet been constructed. On the finite joint space taking part in the exchange, this paper composes free evolution, formation exchange, and pattern reconstruction into a unitary cycle, with the exchange matrix elements and the environment preparation given as conditions of the construction. A standing wave exchanges continuously with the environment during its propagation, and the system state at the end of the cycle and the outgoing records are determined by the same joint evolution; the environment is refreshed cycle by cycle, and the system continues to evolve under the same map. Retrieval, transit, and transport together determine the maintenance of the standing wave, the change of identity, and the formation of records, while the reverse exchange replenishes formation under the corresponding incident occupation. Record weights are determined, under explicit statistical conditions, by the squared norms of the components of the joint state, and energy and the corresponding conserved charges are balanced between system and environment. Under a specified coherent detection preparation, a conserving cycle realizes position records with controlled error and gives error bounds for the interference visibility; the corresponding finite model gives conditional thermalization during identity maintenance and the equilibrium of formation and exit. Under fixed supply and convergence conditions, the steady total occupation of a linear ensemble equals the product of the production rate and the mean waiting time to the first exit. The evolution of a single standing wave, the occupation of many standing waves, and the statistics of actual records are thereby connected within the same cycle.

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1 Introduction

The formation, motion, decay, and thermalization of matter have each been established[2, 3, 4, 5] on the state space of the Bost–Connes system[1]. Once a standing wave has formed from the background, it keeps exchanging with the background as it propagates; it may keep its original identity, transfer energy to the environment, or change identity by transit through the zero element. The existing descriptions give, separately, the propagation action, the selection structure, and part of the rates of these processes; what remains is to compose the actions into a complete single cycle and to determine how the state at the end of a cycle enters the subsequent evolution.

Matter is a stable standing wave maintained by cycles in a thermal bath[2]. Propagation repeats in units of cycles, and each cycle consists, in order, of free evolution, formation exchange, and pattern reconstruction[4, 5]:

$$\text{free evolution} \longrightarrow \text{formation exchange} \longrightarrow \text{pattern reconstruction}.$$

Free evolution gives each component a translation phase; formation exchange contains the outflow of amplitude to the background and its retrieval, and admits transit and transport; pattern reconstruction superposes linearly, on the lattice sites, the components acted on by the first two steps.

Complete retrieval restores the amplitude of the standing wave, and the position of the reconstruction center is determined by the phase relations among the components. The state of the system at the end of a cycle is the initial state of the next cycle.

In formation exchange the outflow component is carried by the environment mode currently taking part in the exchange, and it remains in the joint evolution until the present round trip is complete. Amplitudes leading to the same joint final state are superposed at the amplitude level; the component returning to the system and the component left in the outgoing environment together determine the outcome of the cycle. The formation of a standing wave requires the incident environment to supply the corresponding energy and charge, while the maintenance of a formed standing wave relies on retrieval within the cycle. The environment modes are refreshed cycle by cycle: the incident mode is uncorrelated with the system, and the outgoing mode does not return[3]. Under the same incident conditions the state of the system evolves cycle by cycle under the same cycle action. Exchange changes the occupation of the system, and the outgoing records carry the corresponding transfers; thermalization describes the relaxation of the occupation distribution of the system toward equilibrium under a given background.

This paper constructs a cycle map on the finite subspace taking part in the exchange. The existing no-record single-cycle transfer is retained; outflow, transit, retrieval, and transport are composed on the joint space of system and environment, and pattern reconstruction is carried out by linear superposition. The joint evolution is a unitary action; tracing out the outgoing environment at the end of the cycle gives a completely positive trace-preserving map of the system, and refreshing the environment cycle by cycle makes the dynamics the iteration of this map. When transit and transport are closed, outflow and retrieval are mutually inverse and the cycle recovers the existing no-record propagation; when these branches are open, the no-record share at the end of the cycle and the shares of all outgoing records are determined by the matrix elements of the complete cycle. The transit matrix and the transport couplings are inputs of the construction, and the matching of energy and charge gives the corresponding conservation conditions. Extending the finite description to the complete state space still requires verifying the compatibility of the energy operator with the exchange actions.

2 Cycle Exchange

A standing wave completes one cycle between two successive pattern reconstructions. Free evolution gives each component a translation phase; formation exchange contains the outflow of amplitude to the background, the transit within the background, and the subsequent retrieval; the components acted on by these two steps are superposed again on the lattice sites, which completes the pattern reconstruction[2, 4, 5]. At the end of the cycle the outgoing environment mode no longer takes part in the subsequent evolution, and a fresh incident mode takes over in the next cycle[3].

The map of one cycle must contain the whole amplitude of the present exchange. The outflow component remains in the joint evolution until it is retrieved; only after the cycle is complete are the outgoing environment and the system state entering the next cycle separated.

2.1 A Single Cycle

The system and the environment taking part in the present exchange complete one cycle together. Let $\mathcal{H}$ be the state space of the system, $\mathcal{H}_B$ the space of environment modes, H_B the energy operator of the environment, and

$$\mathcal{H}_{\text{tot}} = \mathcal{H} \otimes \mathcal{H}_B$$

the joint space. This section takes a uniform incident background; couplings and preparations under a non-uniform background must be specified separately. At the start of the n -th cycle the system is in the state ρ_n and the incident environment is in the KMS state ω_β of inverse temperature β [6]. The condition of environment refreshing requires the incident mode to be uncorrelated with the system, so the joint initial state is

$$\rho_n \otimes \omega_\beta.$$

A specified nonthermal detection preparation replaces ω_β by the corresponding environment state ω_B , still requiring the incident environment to be uncorrelated with the system. The joint evolution, the reduction, and the record formulas below apply to this replacement; the invariance of the same-temperature Gibbs state is used only for the corresponding same-temperature preparation. This paper uses natural units $\hbar = c = 1$. The period of the cycle is $\tau = 1/v$, where v is the cycle rate shared by all excitations[2, 4]; the factor $\hbar$ is restored when converting to seconds.

The construction is carried out first on the finite subspace taking part in the exchange. Let $|a\rangle$ be a normalized eigenmode, the label a comprising the particle label, the momentum, and the propagation branch of the mode. The background state is written $|z\rangle$; it is orthogonal to these eigenmodes, and its reference energy level is taken to be zero. This background state belongs to the effective description of the participants in the exchange; its correspondence with the zero element of the complete valuation tower must be specified together with the energy operator.

When no record is left, the single-cycle transfer of the standing wave is given by reference [4]. On the doublet of momentum k , the factors of free evolution and formation exchange are

$$S_\alpha(k) = e^{ik\delta\sigma_z}, \quad C_\alpha = e^{i(m_\alpha/v)\sigma_x}, \quad \delta = \frac{1}{v},$$

where α labels the existing identity of the particle, m_α is its mass, and σ_x, σ_z are the Pauli matrices on the doublet. With the actions on all components collected into S and C , the existing no-record single-cycle transfer is

$$U_0 = CS.$$

On the background state and on the orthogonal complement not taking part in this propagation, the corresponding identity extension is taken.

Let H be the energy operator of the system. The construction uses the common eigenmodes of H and U_0 ,

$$H|a\rangle = E_a|a\rangle, \quad U_0|a\rangle = e^{i\varphi_a}|a\rangle.$$

The energy E_a is identified through the mass spectrum and the dispersion relation of motion[2, 4]; the cycle phase φ_a also carries the sign of the propagation branch. On the chosen positive-frequency branch $\varphi_a = E_a/v$, and the eigenphase of the other propagation branch has the opposite sign. The exchanges below are balanced by energy differences, while propagation keeps the phase of each branch.

The round trip between the standing wave and the background consists of outflow, retrieval, and possibly transit and transport. Let the surviving amplitude of the mode a be r_a , written as

$$r_a = |r_a|e^{i\theta_a}, \quad X_a = 1 - |r_a|^2.$$

The construction of this section is carried out on the sector $0 < |r_a| < 1$; the single-station r_a is identified through the resonance ratio of the corresponding direction, and the origin of this identification is explained in the next subsection.

In the outflow the system passes from the energy level E_a into the background state, and the environment gains the same amount of energy together with the corresponding conserved charge.

Let $|\text{vac}\rangle$ be the environment vacuum and $|1_a\rangle$ the normalized environment state carrying these quantities, and define the two joint states

$$e_{1a} = |a\rangle|\text{vac}\rangle, \quad e_{2a} = |z\rangle|1_a\rangle.$$

Their total energies are equal. The environment states used for different modes are taken to be orthogonal, so these two-dimensional subspaces are mutually orthogonal.

The action of the outflow on each two-dimensional subspace is taken to be

$$F_a = e^{i\theta_a} \begin{pmatrix} |r_a| & i\sqrt{X_a} \\ i\sqrt{X_a} & |r_a| \end{pmatrix}.$$

The two columns are normalized and orthogonal because $|r_a|^2 + X_a = 1$, so F_a is unitary. Define F as the direct sum of these blocks, with the identity on the remaining orthogonal complement. Its action on the incident standing wave is

$$F e_{1a} = r_a e_{1a} + i e^{i\theta_a} \sqrt{X_a} e_{2a}.$$

The first term is the surviving component, the second the outflow component carried by the current environment mode. The phase factor of the outflow component relative to the surviving component is taken to be i ; retrieval uses the inverse rotation $F^\dagger$ under this phase convention.

Transit acts on the outflow component. Let $|b\rangle|\nu\rangle$ be an admissible outgoing joint state, where $|b\rangle$ is the final-state eigenmode and $|\nu\rangle$ carries the emitted pair, with energy $\omega_\nu \geq 0$. Let q_a, q_b, q_ν be the conserved charges of the incident mode, the final-state mode, and the environment state, respectively. Only matrix elements satisfying

$$E_a = E_b + \omega_\nu, \quad q_a = q_b + q_\nu$$

take part in transit. The q here include only those charges whose conservation has been verified in the process and joint state space used; additive physical charges are balanced by ordinary addition, and labels of finite abelian groups by addition within the group. The generation charge is the readout label of the doublet position, and a quark transit can change the generation through a locked link, so the generation charge cannot be entered into the above matching as a general conserved charge; nor is mirror parity taken as a universally conserved charge. The specific selection relations follow reference [5] and are taken up in §5.1. The dimensionless self-adjoint generator of transit is taken to be

$$K_{\text{mid}} = \sum_{a,b,\nu} (\eta_{b\nu a} |b, \nu\rangle \langle z, 1_a| + \overline{\eta_{b\nu a}} |z, 1_a\rangle \langle b, \nu|),$$

with the corresponding action

$$T = e^{-iK_{\text{mid}}}.$$

The sum runs only over admissible outgoing states. The coefficients $\eta_{b\nu a}$ contain the integral of the coupling of this segment over its duration. Reference [5] provides the selection structure and the conditions used in the rate evaluation; these conditions do not yet determine all matrix elements uniquely, and the present construction takes a transit matrix satisfying them as an input. Different channels may share intermediate states, so the overall action is given by the exponential above.

Transport, a single collision in the sense of reference [3], is written $Y = e^{-iK_{\text{tr}}}$, where K_{tr} is a self-adjoint operator composed of energy-matched terms that change the shell or the momentum; its explicit form is given in §2.3.

The present construction performs retrieval after transit and transport after retrieval. This order specifies one concrete finite realization. The net action of the round trip with the background is

$$\boxed{\mathcal{X} = YF^\dagger TF.} \quad (1)$$

Each factor has been defined on the joint space, so $\mathcal{X}$ is unitary. When transit and transport are closed,

$$T = Y = \mathbf{1}, \quad \mathcal{X} = F^\dagger F = \mathbf{1}.$$

Outflow and retrieval then restore the standing wave completely, and the existing single-cycle transfer U_0 is retained.

The exchange takes place after free evolution. Expressing the same round-trip action on the modes after free evolution gives

$$\mathcal{X}_F = (S \otimes \mathbf{1}_B) \mathcal{X} (S^\dagger \otimes \mathbf{1}_B).$$

The complete action of formation exchange is defined as

$$U_X = (C \otimes \mathbf{1}_B) \mathcal{X}_F.$$

Applying free evolution and formation exchange in succession gives

$$\boxed{U_X(S \otimes \mathbf{1}_B) = (U_0 \otimes \mathbf{1}_B) \mathcal{X}.}$$

The left-hand side is written according to the two action steps of the cycle, while the right-hand side combines the given single-cycle transfer with the present round trip to the background. The two sides are equal without any requirement that these factors commute.

Pattern reconstruction composes the components after the actions linearly into amplitudes on the lattice sites. Let $\mathcal{F}$ be the unitary transformation from the position representation to the momentum representation above; the joint action of one cycle in the same position representation is

$$\boxed{U = (\mathcal{F}^{-1} \otimes \mathbf{1}_B)(U_0 \otimes \mathbf{1}_B) \mathcal{X} (\mathcal{F} \otimes \mathbf{1}_B).} \quad (2)$$

The representation transformations at both ends make the input and output of the cycle belong to the same system space. By the unitarity of each factor,

$$U^\dagger U = \mathbf{1}.$$

At the end of the cycle, once the outgoing environment has been traced out, the single-cycle map of the system has the joint-space representation[7]

$$\boxed{\mathcal{E}_\beta(\rho) = \text{Tr}_B \left[U(\rho \otimes \omega_\beta) U^\dagger \right].} \quad (3)$$

If

$$\omega_\beta = \sum_j \pi_j |j\rangle \langle j|,$$

and $\{|r\rangle\}$ is a complete orthonormal basis of environment final states, then

$$J_{rj} = \sqrt{\pi_j} \langle r | U | j \rangle, \quad \mathcal{E}_\beta(\rho) = \sum_{r,j} J_{rj} \rho J_{rj}^\dagger.$$

Joint unitarity gives

$$\begin{aligned}\sum_{r,j} J_{rj}^\dagger J_{rj} &= \sum_j \pi_j \langle j | U^\dagger \left(\mathbf{1} \otimes \sum_r |r\rangle \langle r| \right) U | j \rangle \\ &= \sum_j \pi_j \mathbf{1} = \mathbf{1}.\end{aligned}$$

Hence $\mathcal{E}_\beta$ is completely positive and trace-preserving; the finite operator sum above is the standard representation of a completely positive state change[8, 9].

When the environment state and the couplings are the same cycle after cycle, the fresh incident mode restores the same starting condition, and the dynamics of the system is

$$\boxed{\rho_{n+1} = \mathcal{E}_\beta(\rho_n), \quad \rho_n = \mathcal{E}_\beta^n(\rho_0).}$$

The joint evolution keeps the whole amplitude within the cycle, and environment refreshing lets this system map be repeated cycle after cycle. This environment setting belongs to the description of repeated interactions[10].

2.2 Free Evolution

Free evolution propagates along the lattice and exchanges no amplitude with the background[4, 5]. The geometric lattice on the principal-unit layer is equally spaced in the logarithmic metric, with lattice spacing $\delta = 1/v$. Translation produces a phase when acting on a momentum component, and the involution gives the two orientations opposite phases, so free evolution on each doublet is

$$S_\alpha(k) = \begin{pmatrix} e^{ik\delta} & 0 \\ 0 & e^{-ik\delta} \end{pmatrix}.$$

This is the S used in §2.1. It acts component by component, changes the modulus of no component, and produces no outgoing record.

The surviving amplitude used in formation exchange is taken from the existing resonance ratio. For an odd prime p , let $\chi_s(a)$ be a character function on the unit group $(\mathbb{Z}/p\mathbb{Z})^\times$ and $|\delta_a\rangle$ the orthonormal position vectors. The normalized direction state is

$$|\chi_s\rangle = \frac{1}{\sqrt{p-1}} \sum_{a \in (\mathbb{Z}/p\mathbb{Z})^\times} \chi_s(a) |\delta_a\rangle.$$

Function values and direction vectors are used separately under this convention; when a position indicator function is written as a normalized vector, the normalization factor of the corresponding inner product is used. The convolution operator is diagonal in the multiplicative character basis; the operator normalized at the amplitude-level evaluation point is written $\hat{\mathcal{K}}$, and its eigenvalues are the resonance ratios[2],

$$\hat{\mathcal{K}}|\chi_s\rangle = R_s |\chi_s\rangle, \quad R_s = \frac{L(\frac{1}{2}, \chi_s)}{\zeta(\frac{1}{2})}.$$

Following the propagation evaluation of the formation process[2], the present construction takes the surviving amplitude of a single-station round trip to be R_s . On the corresponding modes,

$$r_a = R_s, \quad X_a = 1 - |R_s|^2.$$

The resonance ratio keeps the fixed amplitude-level evaluation point $1/2$, which is used separately from the actual inverse temperature β of the present incident environment; the construction keeps this amplitude-level evaluation point and does not let it change with the incident temperature. This identification fixes the amplitudes of outflow and retrieval. For multi-station formation the corresponding surviving actions must be applied station by station along the existing path; the R_s of a single direction cannot stand in for the whole path.

On the joint space F contains the surviving and the outflow component; taking the matrix element in the environment vacuum gives exactly the single-station surviving action:

$$\langle \text{vac} | F | \text{vac} \rangle | a \rangle = r_a | a \rangle.$$

The contractive action of $\hat{\mathcal{K}}$ is therefore, in this construction, a partial action inside formation exchange. The no-record share at the end of the cycle and the shares of outgoing records are determined by the composite action of transit, retrieval, and transport.

The argument of the resonance ratio and the cycle phase are given by different evaluations. The former is

$$\theta_a = \arg r_a,$$

the latter, on the positive-frequency branch at rest,

$$\varphi_a(0) = \frac{m_\alpha}{v}.$$

The evaluation of mass contains the complete formation path, the endpoint normalization, and the generational projection[2]. In this construction $F^\dagger$ undoes the action of F through retrieval, and the mass phase is carried by the given formation-exchange factor C . The two are composed within the same cycle, and each evaluation keeps its original meaning.

Free evolution thus provides the translation phase, and the resonance ratio fixes the surviving and outflow parts inside formation exchange. Whether the standing wave is maintained after the cycle depends on the subsequent retrieval and on the other branches.

2.3 Formation Exchange

Formation exchange lets the standing wave complete one round trip with the background. Retrieval keeps the original standing wave, transit sends amplitude to another final state, and transport changes the shell or the momentum. They proceed in the same joint evolution, and the distinction among final states is carried by the environment states left behind.

With the zero element added to the unit-group section above, the structure of merging into the background and emerging from it is described by the zero-element projection[5]. With the normalized position vectors $|\delta_a\rangle$, write

$$D_F = |\delta_0\rangle\langle\delta_0|, \quad (T_a f)(x) = f(x + a),$$

so that

$$[D_F, T_a] = |\delta_0\rangle\langle\delta_a| - |\delta_{-a}\rangle\langle\delta_0|.$$

The two terms respectively absorb amplitude into the zero element and emit it from the zero element. Summing along the direction χ_s gives the vertex of that direction,

$$V_s = \frac{1}{\sqrt{p-1}} \sum_{a \in (\mathbb{Z}/p\mathbb{Z})^\times} \chi_s(a) [D_F, T_a].$$

By the definition of the normalized character vector and the substitution $a \mapsto -a$,

$$\sum_a \chi_s(a) \langle \delta_a | = \sqrt{p-1} \langle \bar{\chi}_s |, \quad \sum_a \chi_s(a) | \delta_{-a} \rangle = \chi_s(-1) \sqrt{p-1} | \chi_s \rangle.$$

Hence

$$\boxed{V_s = | \delta_0 \rangle \langle \bar{\chi}_s | - \chi_s(-1) | \chi_s \rangle \langle \delta_0 |.}$$

The extension of the character at the zero element is taken to be zero, so the vertex has no first-order matrix element between directions. Changing direction must go through the composition of absorption and emission, and the selection structure agrees with reference [5]. The vertex of this unit-group section fixes the entrance and the exit; the complete emission process further requires the final states, environment states, and matrix elements of §2.1.

When the section vertex enters the joint exchange it must correspond to the effective eigenmodes taking part in the exchange. Let $\bar{s}$ be the index of the conjugate direction, $\chi_{\bar{s}} = \overline{\chi_s}$, and let $\Pi_s^\chi = | \chi_s \rangle \langle \chi_s |$ be the projection onto the section direction. From $D_F | \chi_s \rangle = 0$ and $\chi_s(-1)^2 = 1$,

$$D_F V_{\bar{s}} \Pi_s^\chi = | \delta_0 \rangle \langle \chi_s |, \quad -\chi_t(-1) \Pi_t^\chi V_t D_F = | \chi_t \rangle \langle \delta_0 |.$$

For a chosen eigenmode a of direction s , specify an isometric identification ι_a on the two-dimensional subspace such that

$$\iota_a | \delta_0 \rangle = | z \rangle, \quad \iota_a | \chi_s \rangle = | a \rangle;$$

on the orthogonal complement of this two-dimensional subspace ι_a is extended by zero. The final-state mode b of direction t likewise uses ι_b . The joint matrix units used by outflow and by transit are then, respectively,

$$\begin{aligned} (\iota_a D_F V_{\bar{s}} \Pi_s^\chi \iota_a^\dagger) \otimes | 1_a \rangle \langle \text{vac} | &= | z, 1_a \rangle \langle a, \text{vac} |, \\ (-\chi_t(-1) \iota_b \Pi_t^\chi V_t D_F \iota_b^\dagger) \otimes | \nu \rangle \langle 1_a | &= | b, \nu \rangle \langle z, 1_a |. \end{aligned} \tag{4}$$

The environment factors carry the energy and charge specified in §2.1, and the matrix units together with their adjoints enter the corresponding outflow block and the transit generator; the coupling strengths are still fixed by the specified exchange. These local identifications are for the effective eigenmodes; they are not the coarse position embedding $\mathcal{J}_K$ of Appendix A.1, nor do they by themselves give the intertwining relation of the complete energy operator.

Step 1, retrieval. The outflow component and the surviving component lie in the same total-energy subspace. With transit and transport closed, the net action inside the exchange is

$$F^\dagger F = \mathbf{1}.$$

Hence

$$U | \psi \rangle | \text{vac} \rangle = U_{\text{pos},0} | \psi \rangle | \text{vac} \rangle, \quad U_{\text{pos},0} = \mathcal{F}^{-1} U_0 \mathcal{F}.$$

The environment returns to the vacuum and the standing wave continues the existing no-record propagation. For a positive-frequency mode at rest,

$$U_{\text{pos},0} | a \rangle_{\text{pos}} = e^{im_\alpha/v} | a \rangle_{\text{pos}}, \quad | a \rangle_{\text{pos}} = \mathcal{F}^{-1} | a \rangle.$$

Outflow and retrieval restore the amplitude, and the cycle leaves behind the existing mass phase.

This restoration relies on keeping the outflow component within the same cycle. If only the environment-vacuum component were kept before retrieval, the product of the two partial actions would be

$$\langle \text{vac} | F^\dagger | \text{vac} \rangle \langle \text{vac} | F | \text{vac} \rangle | a \rangle = | r_a |^2 | a \rangle,$$

whereas the vacuum matrix element of the complete round trip is

$$\langle \text{vac} | F^\dagger F | \text{vac} \rangle |a\rangle = |a\rangle.$$

The difference is exactly the outflow component discarded midway. Retrieval within the cycle must be computed from the joint evolution.

Step 2, transit. Consider first an isolated transit block with transport closed. Write

$$e_1 = |a, \text{vac}\rangle, \quad e_2 = |z, 1_a\rangle, \quad e_3 = |b, \nu\rangle;$$

the three joint states are orthogonal and have equal total energy; $|\nu\rangle$ is orthogonal to the vacuum, e_3 lies in the subspace on which the global F acts as the identity, and the other transit channels are closed in this computation. Absorbing the phase of the transit coefficient into the definition of e_3 , take

$$T = \exp[-i\eta(|e_3\rangle\langle e_2| + |e_2\rangle\langle e_3|)],$$

where η is the dimensionless transit angle. With $c = |r_a|$ and $X = 1 - c^2$,

$$F e_1 = e^{i\theta_a}(c e_1 + i\sqrt{X} e_2), \quad T e_2 = \cos \eta e_2 - i \sin \eta e_3.$$

The retrieval $F^\dagger$ acts only on the subspace spanned by e_1, e_2 . Multiplying in sequence gives

$$\boxed{F^\dagger T F e_1 = (c^2 + X \cos \eta) e_1 + ic\sqrt{X}(\cos \eta - 1) e_2 + e^{i\theta_a} \sqrt{X} \sin \eta e_3.} \quad (5)$$

All components of the transit block are thereby determined. Their shares are

Component at the end of the cycle	Share
Standing wave in the original eigenmode, environment back in the vacuum	$(c^2 + X \cos \eta)^2$
Emitted pair left in the outgoing environment	$X \sin^2 \eta$
Unretrieved single quantum left in the outgoing environment	$c^2 X (1 - \cos \eta)^2$

The three sum to one. Expanding with $c^2 + X = 1$,

$$\begin{aligned} (c^2 + X \cos \eta)^2 + X \sin^2 \eta + c^2 X (1 - \cos \eta)^2 \\ = c^4 + 2c^2 X + X^2 = 1. \end{aligned}$$

A realization with a fixed retrieval angle therefore in general leaves both an emitted pair and an unretrieved single quantum. Both terms must be counted in the complete record; the share of the emitted pair alone cannot replace the whole loss.

Following the cycle-eigenvalue notation of reference [5], on the branch $0 \leq \eta \leq \pi/2$ the no-record eigenvalue of this realization is

$$\Lambda_a = e^{i\varphi_a}(c^2 + X \cos \eta),$$

and the loss share per cycle is

$$\boxed{\varepsilon_a = 1 - |\Lambda_a|^2 = X \sin^2 \eta + c^2 X (1 - \cos \eta)^2.} \quad (6)$$

For small transit angles the second term begins at fourth order,

$$\varepsilon_a = X \sin^2 \eta + O(\eta^4).$$

The transit share of reference [5] can serve as a matching condition within this range; fixing the complete matrix elements must further conform to all allowed and forbidden outgoing records.

The reverse action is given by the adjoint term of the same self-adjoint generator. From the matrix above,

$$\left| \langle e_1 | F^\dagger T F | e_3 \rangle \right|^2 = X \sin^2 \eta.$$

When the incident environment occupies the emitted-pair mode, amplitude can therefore return from the final-state direction to the initial-state direction. The reverse injection needed for formation and the transit belong to the same joint evolution; whether injection occurs depends in addition on the occupation of the incident environment. When several transit channels share intermediate states, the complete exponential of K_{mid} must be used, and the channel-by-channel two-dimensional rotations no longer hold separately.

Step 3, transport. Transport changes the energy and momentum of the system through the exchange of environment quanta. Let $B_{\lambda,\omega}$ be an operator of the system lowering the energy by $\omega > 0$,

$$[H, B_{\lambda,\omega}] = -\omega B_{\lambda,\omega},$$

where λ labels the corresponding shell or position channel. The environment creation operators satisfy

$$[H_B, b_{\lambda,\omega}^\dagger] = \omega b_{\lambda,\omega}^\dagger.$$

The dimensionless generator of the transport segment is taken to be

$$K_{\text{tr}} = \sum_{\lambda,\omega} \left(\xi_{\lambda,\omega} B_{\lambda,\omega} \otimes b_{\lambda,\omega}^\dagger + \overline{\xi_{\lambda,\omega}} B_{\lambda,\omega}^\dagger \otimes b_{\lambda,\omega} \right).$$

The coefficients $\xi_{\lambda,\omega}$ contain the integral of each coupling over its duration. Nonzero Bohr frequencies are listed here; if equal-energy transport takes part in the cycle, the components with $[H, B_{\lambda,0}] = 0$ are paired with the corresponding zero-frequency environment operators and added to the self-adjoint generator. Each term satisfies

$$[H \otimes \mathbf{1} + \mathbf{1} \otimes H_B, B_{\lambda,\omega} \otimes b_{\lambda,\omega}^\dagger] = 0.$$

Absorption is carried by the adjoint terms.

Shell transport (the shells being the valuation shells of reference [3]) is taken from the adjacent-shell shift[3]. Let V be the shell-raising operator used, defined separately from the section direction vertex V_s . When the shell sector used satisfies

$$[H, V] = (\log p)V,$$

raising a shell is paired with the annihilation of a quantum of frequency $\log p$, and lowering with creation. For a more general common space the shift must first be decomposed according to the actual energy operator. For instance, for a given operator B its component lowering the energy by ω is

$$B_\omega = \sum_E \Pi_{E-\omega} B \Pi_E,$$

where Π_E are the energy projections of H . The frequencies and spectrum of the existing shell chain can be inherited directly only when this decomposition is compatible with the original shell description.

Position transport is carried by matrix elements that change the momentum. Fixing the particle label and the propagation branch, and abbreviating the corresponding energy eigenmode as $|k\rangle$, in the energy-resolved description

$$B_{k'k} = |k'\rangle\langle k|, \quad \omega = E(k) - E(k').$$

The corresponding environment quantum carries the momentum difference as well. The admissible frequencies, momenta, and couplings are determined by the spectrum of the medium; they are inputs of position transport. The coherent combination in position detection must be constructed together with the detection reference; the finite realization and its error are given in Appendix A.2.

These couplings enter the same U together with transit. Since the segments in general do not commute, the shares of the complete cycle are obtained from the matrix elements after composition, and the shares obtained for the segments separately cannot simply be added. The outgoing environment may also keep composite records, whose number and type must be determined by the actual final states.

Every term of the outflow block, the transit block, and the transport is balanced by total energy. Hence

$$[\mathcal{X}, H \otimes \mathbf{1} + \mathbf{1} \otimes H_B] = 0.$$

Since U_0 commutes with H ,

$$[(U_0 \otimes \mathbf{1})\mathcal{X}, H \otimes \mathbf{1} + \mathbf{1} \otimes H_B] = 0.$$

Transforming the energy operator to the position representation at the same time gives

$$\boxed{\begin{aligned} [U, H_{\text{pos}} \otimes \mathbf{1} + \mathbf{1} \otimes H_B] &= 0, \\ H_{\text{pos}} &= \mathcal{F}^{-1} H \mathcal{F}. \end{aligned}} \tag{7}$$

Charge conservation requires outflow, transit, and transport to preserve the same joint charge term by term. The background state is taken to be the corresponding neutral state, and the outflow environment state carries the incident charge; transit is matched as above, and the change of the system charge in a transport term is offset by the environment charge. The two orientations of a doublet carry the same charges of this kind, so S , C , and U_0 preserve them. In neutral transport the system operator preserves the charge and the charge of the environment quantum is zero. Under these conditions charge conservation is given by the same term-by-term matching; finite-group charges use the tensor product of the group representations of system and environment, and one cannot conclude from the symmetry of the system map alone that every record branch preserves the system charge.

Inheriting momentum further requires all final states of the exchange to be balanced together. For incident and outgoing states of definite momentum, transit satisfies $k_a = k_b + k_\nu$, the transport quantum carries $k - k'$, and outflow and retrieval are matched along the same momentum component. If the single-cycle transfer of the chosen realization of motion preserves this momentum operator, and each exchange segment preserves the total momentum of system and environment, then their composition preserves the total momentum. The preservation of the weights of the existing involution blocks gives only the preservation of the distribution of the magnitude of momentum[4], which is not enough to conclude that an arbitrary signed momentum operator satisfies the above requirement.

The exchange of total angular momentum is matched by the joint rotation representation. Let $\mathcal{D}_S(g)$ and $\mathcal{D}_B(g)$ be the rotation representations of system and environment, including orbital and

orientation contributions, with $g \in SU(2)$. In the same representation used by the single-cycle transfer and the exchange, the realization of rotational conservation requires

$$[U_0, \mathcal{D}_S(g)] = 0, \quad [A, \mathcal{D}_S(g) \otimes \mathcal{D}_B(g)] = 0,$$

where A stands for F, T, Y in turn and the conditions hold for every g . $F^\dagger$ then preserves the same representation, and so does the product of all factors; after transforming the system representation at the same time, the complete U therefore has joint rotational symmetry. Its continuous generators give the conservation of each component of the total angular momentum. Preserving an environment orientation label or a single intrinsic-axis component does not replace the joint-representation condition, and the specific rotational couplings must still be specified together with the admissible matrix elements.

This proof concerns the complete cycle. The segmented representations of free evolution and formation exchange are not required to commute separately with one and the same fixed matrix form of H ; conservation holds when the input and the output use the same representation.

2.4 Pattern Reconstruction

Pattern reconstruction composes the components after the first two steps into amplitudes on the lattice sites. The components that have passed through free evolution and formation exchange form the pattern by linear superposition; reconstruction itself adds no further coupling between components[4].

Let the system component on some environment final state after the exchange be $\hat{\psi}'(k)$. Its amplitude on the lattice site x is

$$\psi'(x) = (\mathcal{F}^{-1}\hat{\psi}')(x).$$

$\mathcal{F}^{-1}$ is the representation transformation of the linear composition. The moduli and phases of the component amplitudes have already been determined by free evolution and formation exchange; reconstruction gives the spatial pattern according to these components.

When the record branches are closed, $\mathcal{X} = \mathbf{1}$, and the single-cycle propagation reverts to

$$U_{\text{pos},0} = \mathcal{F}^{-1}CS\mathcal{F}.$$

Its eigenphase on each momentum component is given by reference [4]:

$$\cos \varphi_\alpha(k) = \cos(k\delta) \cos(m_\alpha/v).$$

The present construction therefore keeps the existing dispersion relation of motion in the restriction without transit and without transport.

The reconstruction center is determined by the phase relations among the components. On the propagation branch of phase $-\varphi_\alpha(k)$, a narrow wave packet after n no-record cycles has the form

$$\psi_n(x) = \sum_k A(k) e^{ik(x-x_0) - in\varphi_\alpha(k)} u_\alpha(k),$$

where $A(k)$ is the initial momentum amplitude, concentrated near k_0 , and $u_\alpha(k)$ is the normalized internal vector of that branch. The stationary-phase condition gives the displacement of the packet center,

$$x_n - x_0 \simeq n \frac{\partial \varphi_\alpha}{\partial k}(k_0).$$

The displacement per cycle is thus carried by the existing phase spectrum[4]. Reconstruction makes these components interfere constructively at the new center; neither an additional translation nor a further share of cycle phase is needed.

When the record branches are open, reconstruction acts on the system component corresponding to each environment final state. The partial trace over different record components takes place after the cycle is complete; within one record component the amplitudes are still composed by linear superposition. The distinction that the actual records make among these final states determines the readout of the corresponding events.

2.5 Steady States

A standing wave that keeps its own form after completing a cycle constitutes a steady state. Consider first the restriction to no-record maintenance: the environment is in the vacuum, and the matrix element of the complete cycle on the vacuum final state is written

$$M = \langle \text{vac} | U | \text{vac} \rangle.$$

The vacuum system map is abbreviated here as $\mathcal{E}$. The operators corresponding to the other environment final states are written

$$J_r = \langle r | U | \text{vac} \rangle, \quad r \neq \text{vac}.$$

Joint unitarity gives

$$M^\dagger M + \sum_r J_r^\dagger J_r = \mathbf{1}.$$

Hence M is in general a contraction. For normalized ψ ,

$$1 - \|M\psi\|^2 = \sum_r \|J_r\psi\|^2.$$

The decrease of the no-record share equals the share of all outgoing records.

If the standing wave satisfies

$$M\psi = e^{i\varphi}\psi,$$

its modulus is preserved after one cycle. Substituting into the completeness relation,

$$\sum_r \|J_r\psi\|^2 = 0.$$

Each term is nonnegative, so

$$J_r\psi = 0 \quad \text{for all } r.$$

Hence, when the complete system map acts on this pure state,

$$\mathcal{E}(|\psi\rangle\langle\psi|) = |\psi\rangle\langle\psi|.$$

Conversely, if all record operators annihilate ψ and M merely multiplies it by a phase, the standing wave is maintained cycle by cycle. The condition for a no-record steady state is

$$M\psi = e^{i\varphi}\psi, \quad J_r\psi = 0 \quad \text{for all } r. \tag{8}$$

The first condition prescribes that the form is kept, the second that the present cycle produces no record. A general stationary state of the system is defined by $\mathcal{E}(\rho) = \rho$ and may be accompanied by

environment records that preserve the state; the condition above characterizes only the restriction to no-record maintenance.

In a cycle with transit and transport closed, $M = U_{\text{pos},0}$. The moduli of its propagation eigenmodes are unchanged, and a positive-frequency mode at rest gains the phase m_α/v per cycle, in agreement with the definition of mass[2, 4]. On the isolated transit block of §2.3,

$$M|a\rangle_{\text{pos}} = \Lambda_a|a\rangle_{\text{pos}}, \quad \Lambda_a = e^{i\varphi_a}(|r_a|^2 + X_a \cos \eta_a).$$

With transit closed $|\Lambda_a| = 1$; open and producing loss, $|\Lambda_a| < 1$. With identical fresh environments taking over cycle after cycle, the no-record evolution of the block is

$$M^n|a\rangle_{\text{pos}} = \Lambda_a^n|a\rangle_{\text{pos}},$$

with share

$$|\Lambda_a|^{2n} = (1 - \varepsilon_a)^n.$$

The exponential form is given by the repeated application of the same cycle.

If the no-record action preserves a set of eigenmodes separately,

$$M|a\rangle_{\text{pos}} = \Lambda_a|a\rangle_{\text{pos}},$$

then the superposition $\psi = \sum_a d_a|a\rangle_{\text{pos}}$ satisfies the eigenvalue condition if and only if

$$d_a(\Lambda_a - \Lambda) = 0 \quad \text{for each } a.$$

A nondegenerate eigenvalue therefore corresponds to a single eigenmode; with degeneracy the whole corresponding eigenspace must be kept. Energy conservation alone only ensures that M preserves the energy sectors; whether it is diagonal inside a sector is still decided by the actual exchange matrix.

When the environment is in a general KMS state, the cycle of the system is given by the whole family J_{rj} of §2.1. Different incident occupations may correspond to different no-record actions, and they cannot all be written as one and the same vacuum matrix element M . Equilibrium is then described by the stationarity of the density operator of the system.

In the finite realization, write

$$\sigma_\beta = \frac{e^{-\beta H_{\text{pos}}}}{Z_S}, \quad \omega_\beta = \frac{e^{-\beta H_B}}{Z_B},$$

where Z_S, Z_B are the partition functions of system and environment, normalizing the traces of the corresponding density operators to one. Their product is proportional to the Gibbs weight of the total energy. By the energy conservation of the complete cycle,

$$U(\sigma_\beta \otimes \omega_\beta)U^\dagger = \sigma_\beta \otimes \omega_\beta.$$

Tracing out the environment gives

$$\boxed{\mathcal{E}_\beta(\sigma_\beta) = \sigma_\beta.} \tag{9}$$

The common cycle therefore preserves the same-temperature equilibrium state of the chosen finite energy model. When shell transport is inherited, one must further distinguish the shell-occupation distribution, the position-level distribution, and the Gibbs density operator on the full space[3]: if the shell weights are π_j , the position distribution uniform within a shell is $\mu(x) = \pi_j/|S_j|$; for an energy operator with intra-shell degeneracy, the shell weights of $e^{-\beta H_{\text{pos}}}/Z_S$ also contain the

corresponding degeneracies. Whether the two agree must be verified against the actual energy and exchange model. Whether an arbitrary initial state tends to this equilibrium depends further on the connectivity of the channels and on the coherent structure retained. When the occupations form a closed finite transfer chain, irreducibility and aperiodicity make its equilibrium distribution unique; the cases of shell transport satisfying these conditions are established in reference [3]. If at zero temperature there remain eigenmodes unaffected by records, the coherences among them may continue to evolve, and the relaxation of occupation cannot replace the steady-state condition of the complete state.

In this cycle retrieval maintains the standing wave and the existing phase spectrum moves it; transit and its reverse action connect loss and formation; transport changes the occupation in an environment refreshed cycle by cycle. They all enter the same joint unitary U and the same system map $\mathcal{E}_\beta$. The specific transit matrix, the transport couplings, and their energy matching on the complete state space remain data of the construction; the construction above gives the cycle realization that can be repeated once these data are fixed. The distinguishable outcomes kept by the outgoing environment constitute the records, and the actual readout determines the corresponding events.

3 Records

After a standing wave completes one cycle, the system enters the next cycle in its updated state, and the outgoing environment leaves carrying the products of the present exchange. The change of the system state is determined by the cycle map of §2; how the corresponding process is identified as an event depends on the distinguishable records kept by the outgoing environment. The formation of records, their readout, and their statistical law connect the joint evolution of a single cycle with the events that occur in succession.

3.1 Formation of Records

For an exchange to constitute a record, the environment must still keep the distinction of the present process after leaving the system. The existing record criterion contains three conditions in the readout layer it uses: the distinction is carried by a classical quantum number invariant under propagation and commuting with all observables of that layer; the distinction is carried by a really emitted excitation; the excitation carrying the record escapes and does not return[4]. The emitted pair produced by transit can carry the identity difference, and the outgoing excitation produced by transport can carry the difference of energy and of the other conserved quantities[3, 5].

The outflow component still takes part in the joint evolution until the present round trip is complete. With transit and transport closed, outflow and retrieval satisfy

$$F^\dagger F = \mathbf{1}.$$

The environment returns to its initial state and the standing wave continues to evolve by the existing single-cycle transfer action. With transit and transport open, the final state of the outgoing environment is determined by the complete cycle U . Whether the outflow component can return to the system, and what record the unreturned component carries, must both be decided at the end of the cycle.

The effect of records on the coherence between processes is described by the overlap of the environment final states. Suppose several processes lead to the same normalized system final state $|b\rangle$; label the processes by x , with amplitudes A_x and normalized environment final states $|e_x\rangle$. The

contribution of these processes to the joint final state is

$$|b\rangle \otimes \sum_x A_x |e_x\rangle.$$

The inner product and norm of the joint Hilbert space of §2 are used below. The squared norm of the normalized joint state on a specified final-state component is called its **share**. After the partial trace over the environment, the share of the system final state $|b\rangle$ is

$$w_b = \sum_{x,x'} A_x \overline{A_{x'}} \langle e_{x'} | e_x \rangle. \quad (10)$$

If the processes leave the same environment state, absorbing its phase into A_x gives

$$w_b = \left| \sum_x A_x \right|^2.$$

The environment has kept no distinction between these processes, and the amplitudes superpose coherently. If the environment final states are pairwise orthogonal, the cross terms vanish,

$$w_b = \sum_x |A_x|^2.$$

The shares of the processes can then be added.

Conserved labels provide a stable basis for distinguishing records. The environment subspaces belonging to different conserved charges are mutually orthogonal, and records carrying these charges can be distinguished. Inside the subspace of one and the same conserved charge there may still be differences of energy, momentum, or other state variables, so an equal charge label is not enough to conclude that the environment final states are identical. The coherence between processes is still decided by the actual overlap.

General environment final states allow partial overlap,

$$0 < |\langle e_{x'} | e_x \rangle| < 1,$$

and the corresponding cross terms are partly retained. If the environment differences within one sector are completely eliminated under a given reduction, the effective overlap can further degenerate to zero or one; in the general case the complete overlap matrix must be kept. When an existing reduced evaluation is applied to the present cycle, its compatibility with the actual environment action has to be verified.

The quadratic structure of shares also limits the order of interference terms. For three fixed branch vectors v_1, v_2, v_3 under the same linear action, write

$$w_{\mathcal{A}} = \left\| \sum_{x \in \mathcal{A}} v_x \right\|^2.$$

All shares are computed with the same incident normalization, and the branch vectors do not change with the subset $\mathcal{A}$. When the inner products are expanded, each diagonal term has coefficient $1-2+1$ in the expression below and each cross term has coefficient $1-1$, so

$$w_{123} - w_{12} - w_{13} - w_{23} + w_1 + w_2 + w_3 = 0. \quad (11)$$

This is the quantum-measure relation of vanishing third-order interference[11], valid for linear combinations of fixed branch vectors as above. When a path is actually altered and the evolution of the other branches changes with it, the joint state after the alteration must be used for the computation.

The partition of records is determined by the distinctions the environment actually keeps; the physical determination of the record basis is connected with the stability of the pointer basis under the interaction[12]. The same system map can have different Kraus representations, and this change of representation does not by itself determine the physical records. The readout must rest on the records already formed, and must state which labels are kept and which outcomes are merged.

3.2 Readout of Events

An event is identified by the distinctions kept in the records. When a record is read out, the corresponding system final state and the outgoing environment together constitute the outcome of the present cycle. One outcome may consist of a single excitation, an emitted pair, or a composite record of several excitations.

In the outgoing environment let the orthogonal projections Π_r represent the record subspaces distinguished by the actual readout. The outcome label r includes the outcome in which the selected record is not detected, so that the readout is complete:

$$\Pi_r \Pi_s = \delta_{rs} \Pi_r, \quad \sum_r \Pi_r = \mathbf{1}_B.$$

For an incident system state ρ , the joint state after the cycle is

$$\varrho_{\text{out}} = U(\rho \otimes \omega_\beta)U^\dagger.$$

Taking the outcome together with the corresponding state change gives the description of a quantum instrument[13]. The system action corresponding to outcome r is

$$\mathcal{I}_r(\rho) = \text{Tr}_B[(\mathbf{1} \otimes \Pi_r)\varrho_{\text{out}}(\mathbf{1} \otimes \Pi_r)], \quad (12)$$

with share

$$w_r(\rho) = \text{Tr} \mathcal{I}_r(\rho).$$

Since the cross blocks between different record subspaces vanish under the environment partial trace, summing over all outcomes recovers the system map of §2:

$$\sum_r \mathcal{I}_r(\rho) = \mathcal{E}_\beta(\rho), \quad \sum_r w_r(\rho) = 1.$$

If the readout distinguishes only the union $\mathcal{A}$ of a group of outcomes, then

$$\mathcal{I}_\mathcal{A}(\rho) = \sum_{r \in \mathcal{A}} \mathcal{I}_r(\rho), \quad w_\mathcal{A}(\rho) = \sum_{r \in \mathcal{A}} w_r(\rho).$$

This coarse-graining merges outcomes that have already been distinguished; it does not restore the coherence terms between different record subspaces.

The coherence inside one record subspace must still be determined by the environment states. With the process labels of the previous subsection, the share leading to the system final state $|b\rangle$ and belonging to outcome r is

$$w_{b,r} = \sum_{x,x'} A_x \overline{A_{x'}} \langle e_{x'} | \Pi_r | e_x \rangle.$$

Obtaining the same readout label only says that these processes belong to the same record subspace; whether their amplitudes add coherently depends on the environment overlap within that subspace.

In the restriction of vacuum incidence with all outgoing excitations included in the records, let $r = 0$ denote the no-record outcome. The M and J_r of §2 give the no-record action and the

remaining outgoing actions, respectively. With the environment final states retained, the restriction of the same cycle reads

$$W|\psi\rangle = M|\psi\rangle \otimes |\text{vac}\rangle + \sum_{r \neq 0} J_r |\psi\rangle \otimes |r\rangle, \quad W^\dagger W = \mathbf{1}.$$

For a normalized system initial state $|\psi\rangle$, the completeness relation gives

$$\boxed{1 - \|M\psi\|^2 = \sum_{r \neq 0} \|J_r \psi\|^2 = \sum_{m, r \neq 0} |\langle m | J_r | \psi \rangle|^2.} \quad (13)$$

Here $\{|m\rangle\}$ is a complete orthonormal basis of system final states. The left side is the share by which the no-record component decreases, the middle is the share of all outgoing records, and the right side further distinguishes the system final states corresponding to each record. These expressions sum over the same set of joint final states and must not be counted twice in the total share.

The isolated transit block of §2 contains two outgoing records, the emitted pair and the unretrieved single quantum. With the modulus $|r_a|$ of the surviving amplitude, the outflow share $X_a = 1 - |r_a|^2$, and the transit angle η_a , the shares of the two records are

$$w_{\text{pair}} = X_a \sin^2 \eta_a, \quad w_{\text{single}} = |r_a|^2 X_a (1 - \cos \eta_a)^2.$$

The share of all outgoing records is

$$\varepsilon_a = w_{\text{pair}} + w_{\text{single}} = 1 - (|r_a|^2 + X_a \cos \eta_a)^2.$$

Reading out only the emitted pair gives w_{pair} ; reading out all outgoing excitations completely gives ε_a . The unretrieved single quantum is an independent outgoing outcome and must be kept in the total share.

The physical reading of an event is determined by the conservation relations between system and environment. Let the incident and final-state modes of the system be a, b , the incident and outgoing environment states j, r , with energies $E_a, E_b, \omega_j, \omega_r$ and conserved charges q_a, q_b, q_j, q_r . For matrix elements satisfying the balance conditions,

$$E_a + \omega_j = E_b + \omega_r, \quad q_a + q_j = q_b + q_r.$$

The energy gained by the environment equals the energy lost by the system, and the charge difference of the environment equals the charge transferred by the system; the operation on charges is understood in the conservation group to which they belong. These differences of the conserved quantities constitute the physical readings of transit and transport records; the identities before and after transit must in addition be determined by the corresponding mode labels[3, 5]. In the restriction of shell transport satisfying the energy-matching condition of §2, the energy reading of a single quantum is $\log p[3]$.

The incident occupation of a thermal environment also enters the computation of event shares. If

$$\omega_\beta = \sum_j \pi_j |j\rangle \langle j|,$$

and $\{|r, \mu\rangle\}_\mu$ denotes an orthonormal basis of the record subspace Π_r , then

$$J_{r\mu j} = \sqrt{\pi_j} \langle r, \mu | U | j \rangle,$$

and hence

$$\mathcal{I}_r(\rho) = \sum_{\mu,j} J_{r\mu j} \rho J_{r\mu j}^\dagger.$$

Both the internal label μ and the unmonitored incident index j must be summed over. Only when the incident state is known, or the incident label is also actually kept, can the incident and outgoing readings be compared cycle by cycle to determine the difference of the present exchange. In a general thermal environment the no-response outcome is determined by the specified readout and cannot be identified directly with the outgoing vacuum.

The individual readout outcomes of a single cycle are thus described by $\mathcal{I}_r$ and their shares w_r . Using these shares for event counting requires determining their relation to statistical weights.

3.3 Frequency of Records

Repeated readout of records under the same preparation conditions yields the sample frequencies of the outcomes. The shares given by the joint evolution and the frequencies obtained from a finite number of counts belong to different levels: the former are determined by inner products of the joint state, while the latter fluctuate statistically with the number of repetitions. The two are connected through the statistical weights of the readout outcomes.

The derivation of record weights uses the argument of joint-state swaps and auxiliary fine-graining[14]. For a given record readout the following conditions are adopted:

- (F0) The statistical weights of the outcomes are nonnegative, normalized, and additive over mutually exclusive outcomes.
- (F1) The weights are determined by the joint state and the specified orthogonal readout; under a unitary transformation of the joint space on the outcome side, with the readout projections relabeled by the same transformation, the weights are covariant.
- (F2) A unitary operation on an auxiliary unit already separated from the outcome side does not change the weights of the original outcomes; copying orthogonal labels, finite orthogonal fine-graining of the joint record subspaces, and adding uncoupled auxiliary units preserve the weights of the corresponding coarse outcomes. For a readout that does not act on the purifying reference system, a mixed state and any of its purifications give the same statistical weights.
- (F3) The weights are continuous in the normalized joint state.
- (F4) When the component of the joint state in some outcome subspace is zero, the weight of that outcome is zero.

These conditions are used both for the actual coarse readout and for the finite auxiliary readouts below. In the equal-amplitude argument the outcome side is the system with the outgoing environment, SB ; after auxiliary fine-graining it is SBC' ; the swaps and compensating swaps are understood on the corresponding joint spaces. Joint rank-one branches, together with their orthogonal complements on which the joint state has zero component, are used for the auxiliary argument; the actual detector is not required to have the same resolution. Where relabelings involve different conserved labels, their covariance is adopted as a statistical condition. Environment refreshing keeps the outgoing records separate from subsequent cycles; the conditions above separately prescribe the consistency of the readout under this separation and under the auxiliary extensions. The support condition (F4) excludes outcomes without a corresponding joint-state component; it does not prescribe the weights of nonzero components. The auxiliary fine-graining is finally reduced back to the

actual orthogonal partition of records, and the environment overlaps within a process are still kept according to Eq. (10).

Step 1: outcomes of equal amplitude have equal weights. Consider first a pure joint initial state, and let $|\Psi\rangle$ be the normalized joint state after the cycle. The component corresponding to outcome r is

$$|v_r\rangle = (\mathbf{1} \otimes \Pi_r)|\Psi\rangle.$$

By the orthogonality and completeness of the record projections,

$$\langle v_s | v_r \rangle = 0 \quad (s \neq r), \quad \sum_r \|v_r\|^2 = 1.$$

Write $a_r = \|v_r\|$. The normalized branch

$$|\sigma_r\rangle = \frac{|v_r\rangle}{a_r}$$

is defined only for outcomes with $a_r > 0$. Within each record subspace, further distinguish the one-dimensional space spanned by $|\sigma_r\rangle$ from its orthogonal complement. By (F4) and additivity, the weight of outcome r equals the weight of this nonzero branch.

Copying the record label to an auxiliary unit C gives

$$|\Phi\rangle = \sum_{r:a_r>0} a_r |\sigma_r\rangle_{SB} |\eta_r\rangle_C.$$

Here SB denotes the system with the outgoing environment and $\{|\eta_r\rangle_C\}$ are orthogonal record copies. Suppose $a_r = a_s$; swapping the two branches on SB and swapping the corresponding copies on C restores the joint state:

$$(V_{SB} \otimes V_C)|\Phi\rangle = |\Phi\rangle.$$

By (F1) the swap on the outcome side interchanges the weights; by (F2) the compensating swap on the auxiliary side preserves the original record weights, so $p_r = p_s$. If all nonzero branches have equal amplitude and there are N_* of them, the weights of the remaining outcomes are zero by (F4), and normalization gives

$$p_r = \frac{1}{N_*}.$$

Step 2: rational shares determine the weights through finite fine-graining. Suppose

$$a_r^2 = \frac{m_r}{N_*}, \quad \sum_r m_r = N_*,$$

with positive integers m_r . Introduce an auxiliary unit C' and extend the record copies as follows:

$$|\eta_r\rangle_C |0\rangle_{C'} \mapsto \frac{1}{\sqrt{m_r}} \sum_{\ell=1}^{m_r} |r, \ell\rangle_C |r, \ell\rangle_{C'}.$$

The images for different r are normalized and orthogonal, so this action extends to a unitary action on the auxiliary space. The extended joint state is

$$|\Phi'\rangle = \frac{1}{\sqrt{N_*}} \sum_r \sum_{\ell=1}^{m_r} |\sigma_r\rangle_{SB} |r, \ell\rangle_{C'} |r, \ell\rangle_C.$$

Taking SBC' as the fine-grained outcome side and C as the copy side, the joint state has N_* orthogonal branches of equal amplitude, and the remaining outcomes are excluded by (F4). Each fine outcome has weight $1/N_*$. By additivity and the consistency of fine-graining,

$$p_r = \sum_{\ell=1}^{m_r} \frac{1}{N_*} = \frac{m_r}{N_*} = a_r^2.$$

Step 3: general shares are determined by continuity. Take a sequence of rational share sets, preserving normalization, that approaches an arbitrary set of shares. The weights obtained from each auxiliary fine-graining are first reduced back to the original record space, and the continuous limit is then taken on this fixed space. By (F3),

$$\boxed{p_r = a_r^2 = \|v_r\|^2.} \quad (14)$$

Under the stated readout conditions, therefore, the statistical weight of each outcome equals its share.

The extension to mixed states uses the purification consistency in (F2). Let the mixed joint final state be ϱ ; any two of its purifications are related, after the auxiliary dimensions have been padded to match, by

$$|\Omega'\rangle = (\mathbf{1}_{SB} \otimes V_C)|\Omega\rangle.$$

The original record projections act only on SB and commute with V_C . By the auxiliary unitary invariance in (F2), the two purifications give the same record weights, so the extension does not depend on the choice of purification. From purification consistency and Eq. (14),

$$p_r(\varrho) = \|(\mathbf{1} \otimes \Pi_r \otimes \mathbf{1}_C)\Omega\|^2 = \text{Tr}[(\mathbf{1} \otimes \Pi_r)\varrho].$$

Substituting the joint final state of the cycle,

$$\boxed{p_r(\rho) = \text{Tr}[(\mathbf{1} \otimes \Pi_r)U(\rho \otimes \omega_\beta)U^\dagger] = \text{Tr} \mathcal{I}_r(\rho).} \quad (15)$$

The reference system serves to represent the mixed state; its label need not become an actual reading. In the restriction of vacuum incidence with all outgoing records resolved, $p_0(\psi) = \|M\psi\|^2$ and $p_r(\psi) = \|J_r\psi\|^2$ ($r \neq 0$). Given outcome r with $p_r(\rho) > 0$, the corresponding conditional system state is

$$\rho_r = \frac{\mathcal{I}_r(\rho)}{p_r(\rho)}.$$

The shares of orthogonal records can therefore be added at the probability level, while the coherent contributions retained inside a record are still computed from the actual joint amplitudes. A continuous-time Markovian dissipative master equation can be formulated through conditional histories of no-jump evolution and stochastic quantum jumps[15].

The component that forms no record in a cycle continues to take part in the subsequent cycles. Whether repeated readout is complete depends on whether this share tends to zero as the number of cycles increases. Below, take vacuum incidence, treat any outgoing record as the terminating outcome of the present readout, and distinguish the cycle in which a record is first formed; each cycle uses a fresh environment in the same state, and the outgoing environment of the previous cycle does not return.

If the first $n - 1$ cycles leave no record and the n -th cycle first forms the record r , the corresponding action is

$$D_{r,n} = J_r M^{n-1}, \quad n \geq 1.$$

Different first-record times correspond to different sequences of environment records, and their shares add as separate terms. For at most N repeated cycles, the effect operator corresponding to all first records is

$$Q_N = \sum_{n=1}^N \sum_{r \neq 0} (M^\dagger)^{n-1} J_r^\dagger J_r M^{n-1}.$$

By the completeness relation $\sum_{r \neq 0} J_r^\dagger J_r = \mathbf{1} - M^\dagger M$, the n -th term becomes

$$(M^\dagger)^{n-1} M^{n-1} - (M^\dagger)^n M^n.$$

The terms cancel in succession, giving

$$\boxed{Q_N = \mathbf{1} - (M^\dagger)^N M^N.} \quad (16)$$

For a normalized initial state $|\psi\rangle$, the weight of having formed a record by the N -th cycle is

$$p_{\text{rec}}^{(N)}(\psi) = 1 - \|M^N \psi\|^2.$$

On the subspace under discussion, if $M^N \psi \rightarrow 0$ the total weight of first records tends to one. If there is a no-record eigenmode with $\|M^N \psi\| = \|\psi\|$, prolonging the wait cannot make that mode form a record.

A complete readout of directions further requires that the records identify the incident direction. Below we restrict to the subspace spanned by a set of mutually orthogonal normalized common eigenmodes $|u_k\rangle$. These vectors are all taken in the system position representation of §2; write $P_k = |u_k\rangle\langle u_k|$ and $P_* = \sum_k P_k$, where k runs over the selected readout directions. Require

$$M P_k = P_k M = \Lambda_k P_k. \quad (17)$$

This prescribes that the no-record evolution preserves the direction and excludes other subspaces flowing into that direction through no-record evolution. Denote all outgoing records of direction k by $J_{k\mu}$, with μ distinguishing different records of the same direction, and require

$$J_{k\mu} = J_{k\mu} P_k.$$

These record groups must contain all outgoing outcomes of the selected subspace. Sandwiching the completeness relation with P_k on both sides gives

$$\sum_{\mu} J_{k\mu}^\dagger J_{k\mu} = (1 - |\Lambda_k|^2) P_k.$$

Here Λ_k is the eigenvalue of the no-record action of the selected joint cycle on that direction, and $1 - |\Lambda_k|^2$ is the corresponding record share per cycle. A record can preserve the original identity of the particle, so a decrease of the no-record share does not by itself mean that the identity has ended. After a change of the participating environment or of the joint matrix elements, these conditions must be re-examined against the actual action; a transport coupling alone is not enough to conclude that every direction has a nonzero record share, nor that its outgoing records can distinguish the incident direction.

The action of the first $n - 1$ cycles leaving no record and the n -th cycle first leaving a record of direction k is $D_{k\mu,n} = J_{k\mu} M^{n-1}$. Up to the N -th cycle, the record effect of direction k is

$$Q_{k,N} = \sum_{n=1}^N \sum_{\mu} D_{k\mu,n}^\dagger D_{k\mu,n} = (1 - |\Lambda_k|^2) \sum_{n=0}^{N-1} |\Lambda_k|^{2n} P_k.$$

Hence

$$\boxed{Q_{k,N} = (1 - |\Lambda_k|^{2N})P_k.} \quad (18)$$

If $|\Lambda_k| < 1$, then

$$\lim_{N \rightarrow \infty} Q_{k,N} = P_k.$$

The whole share of a direction is read out jointly by all possible first-record times, and completeness is secured by the no-record share tending to zero with the number of repetitions.

The isolated transit block of §2 provides this decay condition. Within the range of the single-station identification $r_a = R_k$, $0 < |R_k| < 1$, $0 < \eta_k \leq \pi/2$,

$$|\Lambda_k| = |R_k|^2 + (1 - |R_k|^2) \cos \eta_k < 1.$$

If the isolated blocks of different directions can be assembled into a realization that does not mix them, and all of their outgoing environment records can distinguish the incident direction, the complete readout above follows. Both the emitted pair and the unretrieved single quantum must be counted in the record group of each direction. For couplings that share intermediate states or share records, the exclusivity of directions must be checked against the actual matrix elements. The charge difference $q_a - q_b$ carried by a transit record does not by itself always determine the incident direction uniquely, and a general cycle must still compute the outcome distribution with the actual first-record effects.

Identity-preserving records can also complete the readout of stable directions; §6.2 gives the corresponding positive-temperature detection realization. That realization computes the first response from the complete sub-map of the component that has not yet acquired a direction label, and its completeness is given by Eq. (38).

The Born distribution of directions depends on the prepared state and on the subsequent record process. Take the unit-group section $\mathcal{H}_\chi = L^2((\mathbb{Z}/p\mathbb{Z})^\times)$ with the normalized character directions of §2; the unit lattice state satisfies

$$|\delta_1\rangle = \frac{1}{\sqrt{p-1}} \sum_{k=0}^{p-2} |\chi_k\rangle.$$

The correspondence between this section and the complete eigenmodes must be specified separately. Suppose the finite realization used gives an isometric embedding $\mathcal{V} : \mathcal{H}_\chi \rightarrow \mathcal{H}$ with

$$\mathcal{V}^\dagger \mathcal{V} = \mathbf{1}, \quad \mathcal{V}|\chi_k\rangle = |u_k\rangle, \quad |\psi_{\text{in}}\rangle = \mathcal{V}|\delta_1\rangle.$$

The non-principal directions $k \neq 0$ satisfy the no-record direction preservation and the record exclusivity conditions above; the embedding of the principal direction is used only to prescribe the preparation and is not required to complete the same records. The corresponding readout projection is $P_* = \sum_{k \neq 0} P_k$. The embedding must be compatible with the momentum, propagation-branch, and energy description used; the species section cannot be taken directly as the whole propagation space.

Following the surviving preparation inside formation exchange, define in the same system position representation

$$A = \mathcal{F}^{-1} \langle \text{vac} | F | \text{vac} \rangle \mathcal{F}.$$

In the preparation, F is first applied with an independent environment prepared in the vacuum and its final state is read out; the system component corresponding to the vacuum outcome is used for the subsequent evolution. The preparation ends here; the retrieval $F^\dagger$ is not applied to the original

environment, and the remaining outcomes are kept as other preparation outcomes. For a complete basis of environment final states, define

$$A_r = \mathcal{F}^{-1} \langle r | F | \text{vac} \rangle \mathcal{F}, \quad \sum_r A_r^\dagger A_r = \mathbf{1}, \quad A_{\text{vac}} = A.$$

The completeness relation follows from the unitarity of F , so $A^\dagger A \leq \mathbf{1}$, and the share of successful preparation is $\text{Tr}(A \rho A^\dagger)$. The complete cycle then begins with a fresh incident environment, and the outflow, retrieval, and remaining exchanges within the cycle are still computed by the joint action of §2. Below we restrict to realizations in which this action satisfies the contraction condition of §2, and require

$$P_* A (\mathbf{1} - P_*) = 0, \quad A |u_k\rangle = R_k |u_k\rangle \quad (k \neq 0),$$

$$R_k = \frac{L(\frac{1}{2}, \chi_k)}{\zeta(\frac{1}{2})}.$$

The resonance ratio R_k follows the amplitude-level evaluation of the background half-cycle[2]; this paragraph accordingly specifies the spectral values of the surviving component and realizes the preparation by the vacuum selection above. The modes taking part in the spectral evaluation must keep the corresponding coherence relations; the compatibility of this condition on the finite section with the complete energy operator remains to be verified separately, as stated in §2. From the direction expansion of the unit lattice site and the block condition,

$$P_* A |\psi_{\text{in}}\rangle = \frac{1}{\sqrt{p-1}} \sum_{k \neq 0} R_k |u_k\rangle.$$

This vector is kept unnormalized so that the surviving share of the preparation is counted in the outcome weights. Under Eq. (17) and record exclusivity, the weight of detecting direction k by the N -th cycle is

$$p_k^{(N)} = \langle \psi_{\text{in}} | A^\dagger Q_{k,N} A | \psi_{\text{in}} \rangle = \frac{|R_k|^2}{p-1} (1 - |\Lambda_k|^{2N}).$$

Within a finite waiting time, conditional on a non-principal direction having been detected, with nonzero total weight of this condition, the outcome distribution is

$$p_k^{(N), \text{cond}} = \frac{|R_k|^2 (1 - |\Lambda_k|^{2N})}{\sum_{j \neq 0} |R_j|^2 (1 - |\Lambda_j|^{2N})}. \quad (19)$$

Here $1 - |\Lambda_k|^{2N}$ is determined by the repeated action of the same cycle and represents the share of that direction whose record has been completed by the N -th cycle.

If every non-principal direction satisfies the readout conditions above with $|\Lambda_k| < 1$, then in the limit of complete readout

$$p_k^{\text{cond}} = \frac{|R_k|^2}{\sum_{j \neq 0} |R_j|^2} = \frac{|L(\frac{1}{2}, \chi_k)|^2}{\sum_{j \neq 0} |L(\frac{1}{2}, \chi_j)|^2}. \quad (20)$$

This gives the Born distribution under the selected preparation and complete direction readout[2]. The summation is restricted to the records of the non-principal directions; the principal direction and the other preparation outcomes are kept separately, and zero charge does not by itself exclude

other distinguishable records. The record weights of a general cycle are determined by Eq. (15); the directional form of Eq. (20) additionally depends on the preparation and readout conditions above.

If some directions never form records, a longer waiting time can only read out the remaining directions. Under the preparation, direction preservation, and record exclusivity and exhaustiveness conditions above, let the set of non-principal directions with nonzero record share be

$$\mathcal{B} = \{k \neq 0 : |\Lambda_k| < 1\}.$$

By contractivity the other non-principal directions satisfy $|\Lambda_k| = 1$. The efficiency factor in Eq. (18) tends to one inside $\mathcal{B}$ and is identically zero outside. The number of directions is finite, so the numerator and denominator of Eq. (19) can be taken to the limit separately. If $\sum_{j \in \mathcal{B}} |R_j|^2 > 0$, we obtain

$$\boxed{\lim_{N \rightarrow \infty} p_k^{(N), \text{cond}} = \frac{\mathbf{1}_{\mathcal{B}}(k) |R_k|^2}{\sum_{j \in \mathcal{B}} |R_j|^2}.} \quad (21)$$

Here $\mathbf{1}_{\mathcal{B}}(k)$ is the indicator function of the set $\mathcal{B}$. This is the distribution of directions conditional on detection; the directions that form no record remain in the no-record outcome. If the denominator above vanishes, all $p_k^{(N)}$ are zero, no non-principal direction is detected within any finite waiting time, and the limiting conditional distribution is undefined.

The relation between statistical weights and sample frequencies is determined by repeated preparation. Suppose system and environment are prepared anew under the same conditions each time, and the readouts are independent. For outcome r , let the indicator variable of the n -th readout be $Z_r^{(n)}$: one if that outcome is detected, zero otherwise. After N readouts the sample frequency is

$$f_r^{(N)} = \frac{1}{N} \sum_{n=1}^N Z_r^{(n)}.$$

The statistical weight of each readout is p_r , and by independence

$$\mathbb{E}[f_r^{(N)}] = p_r, \quad \text{Var}(f_r^{(N)}) = \frac{p_r(1-p_r)}{N}.$$

For any $\epsilon > 0$,

$$\Pr(|f_r^{(N)} - p_r| \geq \epsilon) \leq \frac{p_r(1-p_r)}{N\epsilon^2}.$$

The sample frequency converges in probability to the statistical weight as the number of repetitions increases, and the fluctuation of a finite sample is described by the variance above. This conclusion uses the record weights already determined; examining the spectrum of a frequency operator alone, or the norm of the branches whose frequency deviates, cannot replace the statistical premise leading from the joint state to the outcome weights[16]. The number of cycles waited for a first record and the number of independently prepared samples are counted separately.

When the same standing wave evolves cycle by cycle, the system initial state of the next cycle is determined by the outcome of the previous one, and the sequence of records is in general correlated. Under the conditions that the incident environment is refreshed independently every cycle and that the cycle action and readout remain the same, the joint weight of a record sequence is

$$\boxed{p(r_1, \dots, r_n \mid \rho_0) = \text{Tr}[\mathcal{I}_{r_n} \circ \dots \circ \mathcal{I}_{r_1}(\rho_0)]}. \quad (22)$$

The conditional weights of the cycles are connected through the system state. Only when these conditional weights do not change with the previous outcomes does the joint weight reduce to the

product of independent repetitions. Summing the unnormalized conditional states over all record sequences recovers the unconditional evolution $\mathcal{E}_\beta^n(\rho_0)$; the corresponding scalar weights sum to one.

The outgoing records thus distinguish the outcomes of a cycle, the complete joint evolution determines the share of each outcome, and the readout conditions turn the shares into statistical weights. Combining these weights with the cycle period further yields the event rates, the evolution law of the identity-maintenance probability, and the time scales of the evolution of occupation distributions.

4 Rates

A standing wave accumulates phase cycle after cycle and exchanges energy and charge with the environment, and the outgoing environment keeps the corresponding records. The record weights determine the share of each outcome per cycle, and the cycle period converts the number of repetitions into time; the identity-maintenance probability and the decay of occupation deviations are determined by iterating the corresponding actions. Below we fix the joint cycle of §2 and the readout of §3, and refresh the incident environment in every cycle with an independent environment mode in the same state, so that the system state satisfies $\rho_n = \mathcal{E}_\beta^n(\rho_0)$.

4.1 Event Counting

One record may contain one or several environment excitations. Counting is done first by the outcomes actually read out. Let $\mathcal{A}$ be the set of selected record outcomes, with action

$$\mathcal{I}_\mathcal{A} = \sum_{r \in \mathcal{A}} \mathcal{I}_r, \quad p_\mathcal{A}(\rho) = \text{Tr } \mathcal{I}_\mathcal{A}(\rho).$$

Let $N_\mathcal{A}^{(N)}$ be the number of times this class of outcome is detected in the first N cycles. The unconditional initial state of the $(n+1)$ -th cycle is ρ_n , and the expectation of the indicator variable is $p_\mathcal{A}(\rho_n)$. Additivity of expectations gives

$$\mathbb{E} N_\mathcal{A}^{(N)} = \sum_{n=0}^{N-1} p_\mathcal{A}(\mathcal{E}_\beta^n(\rho_0)).$$

This relation does not require the successive records of one and the same standing wave to be mutually independent. The cycle period is $\tau = 1/v$, so the mean event-counting rates in the $(n+1)$ -th cycle and over the first N cycles are

$$\mathcal{R}_\mathcal{A}(n) = v p_\mathcal{A}(\rho_n), \quad \overline{\mathcal{R}}_\mathcal{A}^{(N)} = \frac{v}{N} \sum_{n=0}^{N-1} p_\mathcal{A}(\rho_n).$$

For a stationary initial state, $\mathcal{E}_\beta(\rho) = \rho$, this counting rate is constant. If the outcome r resolves the number of excitations n_r it contains, the corresponding excitation-counting rate is $v \sum_r n_r p_r(\rho_n)$.

The cycle rate $v = 246.19 \text{ GeV}$ follows the existing energy-scale evaluation[2, 4]. This paper uses $\hbar = c = 1$; converted to seconds, the cycle period is

$$\tau = \frac{\hbar}{v} \simeq 2.7 \times 10^{-27} \text{ s}.$$

Event counting determines the mean number of records per unit time. How many cycles a standing wave passes through before it first exits its original identity must in addition be computed from all branches that preserve that identity.

4.2 Maintenance and Exit

Transport records can preserve the identity of a particle, so the identity-maintenance probability must include these record histories. Let P be the projection onto the sector of the original particle identity, with an initial state satisfying $P\rho_0P = \rho_0$ and $\text{Tr}\rho_0 = 1$. Denote all Kraus operators of the actual record group that terminates this identity by D_ν ; the remaining outcomes make up the identity-preserving action $\mathcal{Q}$, so that

$$\mathcal{E}_\beta(\rho) = \mathcal{Q}(\rho) + \sum_\nu D_\nu \rho D_\nu^\dagger.$$

This partition requires the selected records to be sufficient to distinguish identity exits from identity-preserving outcomes. If a coarse label contains different identity destinations, the corresponding branch decomposition must be kept.

The identity-preserving evolution stays cycle after cycle in the subspace projected by P . Assume that for every positive operator ρ supported in this sector,

$$\mathcal{Q}(\rho) = P\mathcal{Q}(\rho)P, \quad P\left(\sum_\nu D_\nu^\dagger D_\nu\right)P = \varepsilon P, \quad 0 \leq \varepsilon \leq 1.$$

ε is the exit share per cycle, constant within this sector. By the trace preservation of the complete map,

$$\text{Tr}\mathcal{Q}(\rho) = \text{Tr}\rho - \text{Tr}\left[\rho \sum_\nu D_\nu^\dagger D_\nu\right] = (1 - \varepsilon) \text{Tr}\rho.$$

Let $\tilde{\rho}_n = \mathcal{Q}^n(\rho_0)$; its trace is the probability that the original identity is maintained through the first n cycles. Since $\mathcal{Q}$ preserves the sector, the relation above can be applied cycle after cycle, giving

$$\boxed{P_P(n) = \text{Tr}\tilde{\rho}_n = (1 - \varepsilon)^n.} \quad (23)$$

When this probability is nonzero, $\tilde{\rho}_n/P_P(n)$ is the state conditional on the original identity having been maintained through the first n cycles. This history terminates at the first exit; particles formed anew afterwards are counted separately. If the exit records are exactly all decay records of the particle under consideration, P_P is the non-decay probability; in the restriction where the maintenance branch contains only the no-record action, it corresponds to the survival probability of reference [5].

For $0 < \varepsilon < 1$, the cycle number N_P of the first exit is distributed as

$$\text{Pr}(N_P = n) = \varepsilon(1 - \varepsilon)^{n-1}, \quad n = 1, 2, \dots$$

Measuring the waiting time to the first exit by $T_P = N_P/v$, and summing the tail probabilities,

$$\boxed{\mathbb{E}T_P = \frac{1}{v} \sum_{n=0}^{\infty} \text{Pr}(N_P > n) = \frac{1}{v\varepsilon}.} \quad (24)$$

This gives the mean waiting time to the first exit; under the matching with the actual decay records above, it is the corresponding mean lifetime. On the other hand, at the discrete times $t_n = n/v$, Eq. (23) can be written $P_P(n) = e^{-\Gamma_P t_n}$, where

$$\boxed{\Gamma_P = -v \log(1 - \varepsilon), \quad \tau_{\text{exp}} = \Gamma_P^{-1}.} \quad (25)$$

Γ_P is the exponential rate of the first exit and τ_{exp} the corresponding exponential time scale; at finite ε the latter differs from the mean waiting time. For a small exit share, $\Gamma_P = v\varepsilon + O(v\varepsilon^2)$.

For $\varepsilon = 0$ no exit occurs; for $\varepsilon = 1$ the exit is completed in the readout of the first cycle, and a description by a finite exponential rate no longer applies.

If all records in the selected sector terminate the identity, and $\mathcal{Q}(\rho) = M\rho M^\dagger$ with a no-record eigenmode $M\psi = \Lambda\psi$, then

$$1 - \varepsilon = |\Lambda|^2, \quad \Gamma_P = -2v \log |\Lambda|.$$

A general cycle containing identity-preserving records must still use $\mathcal{Q}$; the first-exit time cannot be computed from the vacuum matrix element M alone.

The isolated transit block of §2 gives a concrete realization of this restriction. Identify the surviving amplitude of a single station in formation exchange by the resonance ratio R_s , with $0 < |R_s| < 1$, and internal outflow share $X_s = 1 - |R_s|^2$. Take $0 \leq \eta_s \leq \pi/2$, close the other transits and transports, and assign all outgoing records to the termination of this identity; then

$$\varepsilon_s = X_s \sin^2 \eta_s + |R_s|^2 X_s (1 - \cos \eta_s)^2.$$

The two terms are the shares of the emitted-pair record and of the unretrieved-single-quantum record. When the two are combined, the geometric law counts the first exit of the tracked identity; identification with the actual decay lifetime requires the corresponding terminating record groups to agree. The conditional transit share is $P_{\text{tr}} = \sin^2 \eta_s$, and for small angles

$$\varepsilon_s = X_s P_{\text{tr}} + O(\eta_s^4).$$

For $\eta_s > 0$ the share of the unretrieved single quantum relative to that of the emitted pair is $|R_s|^2 \tan^2(\eta_s/2)$. The transit angle and the retrieval angle $\gamma_s = \arccos |R_s|$ of outflow and retrieval characterize different actions.

In the narrow-line readout restriction of reference [5], when the existing width is matched to an isolated block with the same definition of decay, so that $\Gamma_P = \Gamma_d$, one has

$$\varepsilon_s = 1 - e^{-\Gamma_d/v}, \quad \cos \eta_s = \frac{e^{-\Gamma_d/(2v)} - |R_s|^2}{X_s}.$$

This branch requires $|R_s|^2 \leq e^{-\Gamma_d/(2v)} \leq 1$. Taking the exchange station of the weak direction, $X_3 = 0.3836$, as an example, the small-share conversion gives

Process	Γ_d [GeV]	$\varepsilon \simeq \Gamma_d/v$	$P_{\text{tr}} \simeq \varepsilon/X_3$
$\mu \rightarrow e\nu\bar{\nu}$	3.01×10^{-19}	1.22×10^{-21}	3.19×10^{-21}
τ (all branches)	2.27×10^{-12}	9.22×10^{-15}	2.40×10^{-14}

X_3 refers to the exchange station on the selected path; the residence direction of the particle itself keeps its existing identification. Multi-station formation paths, node factors, generational projections, and phase space are still treated by the existing evaluation; their matching with the actual cycle requires all outgoing outcomes to be compatible. Maintenance and exit are determined by the branches of one and the same exchange; the relaxation of occupation further involves the unconditional composition of these branches.

4.3 Occupation Relaxation

When the occupation of the system deviates from the stationary state, its decay is decided by the complete map. Let B_a be an eigenmode in the space of density operators,

$$\mathcal{E}_\beta(B_a) = z_a B_a.$$

It is a different object from the eigenmodes in the Hilbert space of §2. Iteration gives $\mathcal{E}_\beta^n(B_a) = z_a^n B_a$; for $0 < |z_a| < 1$ the decay rate of this nonstationary mode is

$$\boxed{\Gamma_{\text{rel},a} = -v \log |z_a|}. \quad (26)$$

If an eigenvalue has a nontrivial Jordan block, the corresponding component also contains a polynomial factor in the number of cycles; the formula above gives its exponential scale. The uniqueness of the stationary state, together with the rest of the spectrum lying strictly inside the unit circle, guarantees that a finite system tends to that stationary state. Trace preservation, environment refreshing, or the invariance of the stationary state does not by itself replace these spectral conditions.

The eigenvalues of the no-record action cannot directly stand in for z_a . Even when M is diagonal, the factor of $M\rho M^\dagger$ on $|\psi_a\rangle\langle\psi_b|$ is $\Lambda_a\overline{\Lambda_b}$, while the complete map still contains all record terms; its spectrum must be computed on the actual operator space or on a closed occupation subspace.

The continuous relaxation rates of reference [3] can be inherited in a compatible weak-collision limit. Fix a finite tower depth K , and suppose the eigenvalue of the slow mode under consideration satisfies the following, with $\Re g_a > 0$:

$$z_a(s) = 1 - s^2 g_a + o(s^2), \quad s^2 \rightarrow 0.$$

Keeping $Ns^2 = u$ fixed,

$$N \log z_a(s) = -Ns^2 g_a + No(s^2) \longrightarrow -u g_a.$$

Hence $z_a(s)^N \rightarrow e^{-u g_a}$. Measuring physical time by $t = N/v$, define the total rate unit

$$\boxed{\kappa = vs^2, \quad \Gamma_{\text{rel},a} = \kappa \Re g_a + o(\kappa)}. \quad (27)$$

Here s^2 is the square of the collision strength of the chosen collision model. This expansion applies to slow modes satisfying the near-identity condition, or to restrictions in which the propagation phase has been explicitly separated; not all modes of a general joint cycle need satisfy it. If the complete reduced action converges under a compatible scaling to a continuous completely positive trace-preserving semigroup, its generator has the Gorini–Kossakowski–Sudarshan–Lindblad form[17, 18].

When the energy calibration and the KMS condition of the shell exchange of reference [3] are inherited, the ascent and descent rates satisfy

$$\frac{\Gamma_\uparrow}{\Gamma_\downarrow} = p^{-\beta}, \quad \kappa = \Gamma_\uparrow + \Gamma_\downarrow.$$

The KMS condition fixes their ratio; the absolute strength is still determined by the transport matrix elements and the environment spectrum. The outflow share X_s inside formation exchange can be completely retrieved and therefore cannot stand in for s^2 . The relaxation of occupation depends on the moduli of the nonstationary eigenvalues of the complete map; the relative phases retained in no-record propagation give the readings of frequencies.

4.4 Phase and Frequency Shift

The relative phase accumulated by eigenmodes over the cycles can be compared by coherent readout. Take orthonormal no-record eigenmodes ψ_a, ψ_b with $0 < |\Lambda_a|, |\Lambda_b| \leq 1$, and write

$$M\psi_a = \Lambda_a\psi_a, \quad \Lambda_a = |\Lambda_a|e^{i\varphi_a}.$$

When the first n cycles leave no record, the unnormalized coherence component satisfies

$$M^n |\psi_a\rangle \langle \psi_b| (M^\dagger)^n = (\Lambda_a \bar{\Lambda}_b)^n |\psi_a\rangle \langle \psi_b|.$$

The relative frequency and the single-mode no-record decay rate are

$$\Omega_{ab} = v(\varphi_a - \varphi_b), \quad \Gamma_{0,a} = -2v \log |\Lambda_a|.$$

The phase is taken on a continuously chosen branch; the discrete cycle determines the frequency only modulo $2\pi v$, and the corresponding reference branch must follow the existing propagation identification. An overall common phase does not change the event count. $\Gamma_{0,a}$ equals the exponential rate of the corresponding first exit only in the restriction of §4.2 in which all records terminate the identity; its identification with the actual decay rate further requires the terminating record groups to agree.

For $j = a, b$, let the reference eigenvalues be $\Lambda_j^{(0)} \neq 0$, changing by $\delta\Lambda_j$ along the continuously corresponding modes. Write $h_j = \delta\Lambda_j/\Lambda_j^{(0)}$, require $|h_j| \ll 1$, and take the same local branch of the logarithm. From

$$\log \Lambda_a - \log \Lambda_a^{(0)} = \log(1 + h_a) = h_a + O(|h_a|^2)$$

we obtain

$$\delta\Omega_{ab} = v \Im(h_a - h_b) + O(v|h_a|^2 + v|h_b|^2), \quad \delta\Gamma_{0,a} = -2v \Re h_a + O(v|h_a|^2).$$

The modulus and the argument of the eigenvalue determine the no-record decay and the phase change respectively. In a degenerate subspace the actual perturbation matrix must be treated first; undetermined reference directions cannot be used one by one.

The phase per cycle of the positive-frequency branch at rest is $m_\alpha/v[2, 4]$. The mass and its vertex frequency shift are taken from the existing evaluation, and their correspondence with the present cycle must be verified through the actual phases. The resonance ratio, the number of equivalent paths, and the Born-level bare coupling used are[2]

$$R_k = \frac{L(1/2, \chi_k)}{\zeta(1/2)}, \quad \alpha_k^{(0)} = \frac{|R_k|^2}{N_k},$$

$$N_k = |\ker \chi_k| m c_k, \quad m = \frac{p-1}{2}, \quad c_k = 1 + [\chi_k = \bar{\chi}_k].$$

m is the generation number and c_k the self-conjugacy factor; N_k counts the indistinguishable equivalent paths, and k is the residence coordinate of the matter family taking part in the resonance. For $p = 7$ the denominators of the three classes are 3, 6, 18. The amplitude-level evaluation point here is fixed at $1/2$, while the actual inverse temperature of the incident environment is β ; the two are used separately.

The corrected-level coupling in a specified process and the factor $(1 - \alpha_{\text{eff}})$ are still defined by the existing evaluation[2]. Let $m_{\alpha, \text{ref}}$ be the reference mass reading before that specified frequency-shift action. If the existing evaluation gives $m_\alpha = m_{\alpha, \text{ref}}(1 - \alpha_{\text{eff}})$, then with the reference branch unchanged the required phase change is

$$\delta\varphi_\alpha = -\alpha_{\text{eff}} \frac{m_{\alpha, \text{ref}}}{v}.$$

It must be matched term by term with the eigenphases of the actual cycle; a mass m_α already identified above is not subjected to the same correction again. The Lamb shift in a weak-coupling environment generator is defined by its environment correlation function and the secular-averaging condition[3, 19]; it cannot be equated with the vertex frequency shift here without matching. Phase and occupation spectra use different readouts, and their time scales are determined by their respective decaying objects.

4.5 Time Scales of Relaxation

The relaxation of shell occupation and of channel structure proceeds on different time scales. Below we first fix a finite K , take a weak-collision generator compatible with the shell chain of reference [3], and then discuss the variation of the tower depth. The shell index is $j = 0, \dots, K$ and the chain length $K + 1$. The nonstationary spectral factors given by uniform ascent and descent rates with reflecting boundaries are

$$g_{\beta,K,r} = 1 - \frac{2p^{\beta/2}}{1+p^\beta} \cos \frac{r\pi}{K+1}, \quad r = 1, \dots, K.$$

The slowest mode of shell occupation corresponds to $r = 1$; in this generator

$$\Gamma_{\text{shell}} = \kappa g_{\beta,K,1}, \quad \tau_{\text{shell}} = \frac{1}{\kappa g_{\beta,K,1}}.$$

As $K \rightarrow \infty$ the spectral gap tends to $1 - 2p^{\beta/2}/(1+p^\beta)$. The value $K = 41$ in the numerical examples below follows the finite-tower closure value of reference [2]. For $p = 7$, $\beta = 1$, $K = 41$, the nonzero rates of the finite chain lie between 0.340κ and 1.660κ ; the corresponding infinite-depth endpoints are about 0.339κ and 1.661κ .

If the actual occupation transfer matrix is exactly $T_{\text{sh}} = I + s^2 L_{\text{sh}}$, where the nonzero eigenvalues of L_{sh} are $-g_{\beta,K,r}$ and the parameters make T_{sh} a stochastic matrix, then the discrete eigenvalues are $z_r = 1 - s^2 g_{\beta,K,r}$; for $0 < |z_r| < 1$ their exponential scale is given by $-v \log |z_r|$, and an eigenmode with $z_r = 0$ disappears after a single action. A general finite cycle inherits the generator rates only within the approximation range of Eq. (27).

Beyond shell occupation, the primitive character moments of the position probability $q_t(x)$,

$$c_\chi(t) = \sum_x \bar{\chi}(x) q_t(x),$$

are the readout of the channel structure. In the position model of reference [3], the primitive character moments of conductor p^K on the unit shell satisfy

$$c_\chi(t) = e^{-\nu_\beta t} c_\chi(0), \quad \nu_\beta = \Gamma_\uparrow = \frac{\kappa}{1+p^\beta}.$$

The channel components of lower conductors p^{K-d} form a $(d+1)$ -dimensional cross-shell tower and cannot each be treated as a single amplitude mode. In the deep-tower limit at $\beta = 1$, the slowest mode becomes asymptotically slower with d as p^{-d} ; the spectrum of a finite tower is given by the eigenvalues of the actual tridiagonal block.

The generation label of the residue field changes by transit through the zero element after reaching the deepest shell. The deep-tower limit at fixed $\beta > 0$ gives the leak rate of the effective superselection[3]

$$\Gamma_{\text{sel}} = \frac{\kappa}{1+p^\beta} (1 - p^{-\beta})^2 p^{-\beta(K-1)} [1 + o(1)], \quad \tau_{\text{sel}} = \Gamma_{\text{sel}}^{-1}.$$

For $p = 7$, $\beta = 1$, $K = 41$ its asymptotic value is $\Gamma_{\text{sel}}/\kappa \simeq 1.44 \times 10^{-35}$. Under the calibration $\kappa = v s^2$, the corresponding time is about 6 years/ s^2 . This is first of all the time scale of the generator used; to take it as the long-term reading of an actual finite cycle, the relative error of this slow mode must be controlled. An absolute remainder that is merely $o(\kappa)$ may exceed the exponentially small Γ_{sel} and does not suffice for this conversion. Transit and the other records may also change the label, and their contributions must be counted separately within the same cycle.

Let $\tau_{\text{char}} = 1/\nu_\beta$ be the relaxation time of the primitive character moments. Within the applicable time window $\tau_{\text{char}} \ll t \ll \tau_{\text{sel}}$, the generation label constitutes an effective superselection quantity. The thermal label leak rate at finite K is nonzero; whether section transit keeps the existing rate depends in addition on the energy and exchange compatibility of Appendix A.1.

Environment refreshing cycle after cycle allows the same reduced map to be iterated; with energy matching and an environment at the same temperature, the σ_β of §2 is a stationary state. The data-processing inequality of relative entropy gives

$$S(\mathcal{E}_\beta(\rho) \parallel \sigma_\beta) \leq S(\rho \parallel \sigma_\beta).$$

This gives monotonicity relative to that stationary state[3, 20, 21]; convergence still depends on occupation closure and on the spectral conditions for coherences. Zero-temperature spontaneous decay is further limited by the incident occupation, and the transit generator itself retains the reverse action. The separation of the time scales and the balance of exchanges between system and environment together determine the evolution of the standing wave.

5 Conservation Laws of the Dynamics

A standing wave keeps its quantum numbers during formation and propagation, and exchanges energy, momentum, and the corresponding charges with the environment during transit and transport. The standing wave that continues to propagate after the exchange and the outgoing environment together carry the incident totals, and the records keep the transferred amounts. What is conserved is determined by the complete process; whether the occupations and labels inside the system are preserved depends in addition on the channels taking part in the exchange.

5.1 Propagation and Exchange

Preservation during propagation and transfer during exchange are joined within the same cycle. The joint action of §2 is the composition of the single-cycle transfer U_0 , the outflow F , the transit T , the retrieval $F^\dagger$, and the transport Y . All comparisons below are made in the same input and output representation; the system energy is abbreviated as $H = H_{\text{pos}}$, and the environment energy remains H_B .

Energy is preserved in free propagation and, in exchange, balanced in terms of the total of system and environment. The single-cycle transfer commutes with the system energy, and every matrix element of outflow, transit, and transport connects joint states of equal total energy, so Eq. (7) gives

$$[U, H \otimes \mathbf{1} + \mathbf{1} \otimes H_B] = 0.$$

The energy before transit is taken over jointly by the final-state eigenmode and the emitted pair[5]; absorption and emission in transport change the energies of system and environment in opposite directions[3].

The correspondence between energy and cycle phase follows reference [4]. Relative to the zero-energy, zero-phase reference chosen in §2, the phase per cycle of the positive-frequency propagation branch is E_a/v , and m_α/v at rest; the phase of the other propagation branch has the opposite sign. The exchange frequency is determined by the actual level difference $\omega = E_a - E_b$. In the restriction where shell labels are compatible with energy, $H = (\log p)N + H_u$, where N is the shell-index operator and H_u the intra-shell energy part; when the shell-raising operator used also satisfies $[H, V] = (\log p)V$, the exchanged quantum of shell transport is taken to be $\log p$. Within the range of this common energy identification and the chosen propagation branch, the energy balance of

formation and decay, the phase readout of motion, and the energy transfer of thermal transport are described by one and the same H .

The preservation of momentum and angular momentum in the single-cycle transfer is given by the involution blocks and their intrinsic axes[4]. Fixing the particle label, let Π_k^{mom} be the projection onto the doublet $\{\pm k\}$, $\hat{e}(k)$ the intrinsic axis, and $\vec{\sigma}$ the Pauli matrices; then

$$[U_0, \Pi_k^{\text{mom}}] = 0, \quad [U_0(k), \hat{e}(k) \cdot \vec{\sigma}] = 0.$$

This single-cycle transfer preserves the distribution of the magnitude of momentum and the angular-momentum component along the intrinsic axis. Exchange transfers momentum and angular momentum between matter and environment; the conservation of total momentum in the complete cycle requires the single-cycle transfer and the exchange segments to satisfy total-momentum matching jointly, and the conservation of total angular momentum requires each segment to preserve the joint rotation representation including orbital and orientation contributions (§2.3).

The preservation of matter directions is carried by the character selection rule. Convolution is diagonal in the character basis, and an allowed vertex keeps the total character of the initial state equal to that of the final state, by character multiplication coordinate by coordinate over the CRT coordinates[2]. Let $\mathbf{G}$ be the finite abelian group used, and $D_S(g)$, $D_B(g)$ the group representations of system and environment. If the single-cycle transfer, outflow, transit, and transport preserve the same joint representation, then retrieval and the composition preserve it as well; transforming the charge representations at both ends simultaneously gives

$$[U, D_S(g) \otimes D_B(g)] = 0 \quad (g \in \mathbf{G}). \quad (28)$$

Between joint states of definite character, the nonzero matrix elements therefore satisfy

$$\prod_{\text{in}} \chi = \prod_{\text{out}} \chi.$$

Labels valued in the finite abelian groups are added within the group to which they belong.

Electric charge and the color representation constrain the admissible transit final states. The charge difference between the two ends of the line is carried by the emitted pair; for instance, the total charge of the final state of $\mu^- \rightarrow \nu_\mu + (e^-, \bar{\nu}_e)$ is $0 - 1 + 0 = -1$, equal to the incident charge. The color constraint is imposed according to the existing vertex identification, in which the colorless intermediate state of β decay restricts the corresponding projection to the color-singlet component[5]. These selection conditions together restrict the nonzero elements of the transit matrix of §2.

The generation charge is carried by the doublet position read out on the lattice. With the notation of the physical identification $p = 7$, take the j -th doublet to be $\{a_j, -a_j\}$ and encode its $\mathbb{Z}/3$ label by $j = 0, 1, 2$,

$$E_j = |\delta_{a_j}\rangle\langle\delta_{a_j}| + |\delta_{-a_j}\rangle\langle\delta_{-a_j}|, \quad G = \sum_{j=0}^2 j E_j.$$

G describes the generation record and is defined separately from the $\mathbb{Z}/3$ character coordinate of the direction[5]. The formation branch preserves the lattice configuration within a family block, and an established generation record is preserved under propagation and under strong and electromagnetic processes[2, 5]. In transit the lepton line continues its generation label and the emitted pair is generation-neutral; the lattice projection of a quark can read out a locked link, the existing link

between the first and second generations having amplitude λ and share λ^2 [5]. The preservation of the generation charge is therefore restricted to the corresponding processes.

The existing decay vertex continues one line and emits one pair; the catalogue used and its admissible compositions preserve the net number of lines. With B the baryon number, this catalogue excludes changes with $\Delta B = 2$ [5]. This restriction enters K_{mid} with the catalogue, while the total number of final-state standing waves may still change.

The protection of the vector current enters the transit coefficients with the existing β -decay evaluation: the strong-dwell insertion used is diagonal in the character basis and commutes with the doubling map μ_2 , so the weight of the vector readout remains one, while the axial-vector readout is not protected by this relation[5].

The symmetry under lattice reflection is established only for the verified convolution action; for the single-cycle transfer and for the physical mirror image it must be judged separately. The mirror asymmetry of weak processes is determined by their winding structure[5]; mirror parity is not listed as a universally conserved charge of the present cycle. The preservation relations and exchange constraints above determine the differences that records can carry.

5.2 Balance of Records

After the exchange is complete, the system that continues to evolve and the environment that leaves between them carry the joint totals. Preservation of the total first gives the balance of the mean changes on the two sides; when the records resolve the corresponding charge values, the same relation further restricts every admissible branch.

For a chosen incident environment state ω_B , abbreviate the system map as

$$\mathcal{E}(\rho) = \text{Tr}_B[U(\rho \otimes \omega_B)U^\dagger], \quad \omega'_B = \text{Tr}_S[U(\rho \otimes \omega_B)U^\dagger].$$

Let Q_S, Q_B be self-adjoint additive quantities on system and environment, whose total

$$Q_{\text{tot}} = Q_S \otimes \mathbf{1} + \mathbf{1} \otimes Q_B$$

satisfies $[U, Q_{\text{tot}}] = 0$. Using the cyclicity of the trace,

$$\text{Tr}[Q_{\text{tot}}U(\rho \otimes \omega_B)U^\dagger] = \text{Tr}[U^\dagger Q_{\text{tot}}U(\rho \otimes \omega_B)] = \text{Tr}[Q_{\text{tot}}(\rho \otimes \omega_B)].$$

Taking the partial traces over system and environment separately gives

$$\boxed{\text{Tr}[Q_S \mathcal{E}(\rho)] - \text{Tr}[Q_S \rho] = \text{Tr}[Q_B \omega_B] - \text{Tr}[Q_B \omega'_B].} \quad (29)$$

The mean amount transferred from the system to the environment equals the mean amount the environment receives from the system. This step does not require the incident environment state or the readout projections to commute with Q_B , and applies to the self-adjoint additive quantities satisfying the joint conservation condition of §5.1.

If the records keep a definite charge value of the environment, the balance can be refined to branches. With the actual record projections Π_r of §3, now take

$$[\omega_B, Q_B] = 0, \quad [\Pi_r, Q_B] = 0.$$

One can then choose an incident common eigenbasis $|j\rangle$ and an outgoing charge eigenbasis $|r, \mu\rangle$ within each record subspace such that

$$\omega_B = \sum_j \pi_j |j\rangle\langle j|, \quad Q_B |j\rangle = q_j |j\rangle, \quad Q_B |r, \mu\rangle = q_{r\mu} |r, \mu\rangle.$$

Here $\pi_j \geq 0$, $\sum_j \pi_j = 1$, and μ only resolves the environment states inside outcome r without adding labels to the actual readout. The corresponding Kraus operators and record actions are

$$J_{r\mu j} = \sqrt{\pi_j} \langle r, \mu | U | j \rangle, \quad \mathcal{I}_r(\rho) = \sum_{\mu, j} J_{r\mu j} \rho J_{r\mu j}^\dagger.$$

Taking the above environment matrix elements of $Q_{\text{tot}} U = U Q_{\text{tot}}$ gives

$$\boxed{Q_S J_{r\mu j} - J_{r\mu j} Q_S = (q_j - q_{r\mu}) J_{r\mu j}.} \quad (30)$$

If the charges of the incident and outgoing system states are q_a and q_b , taking the system matrix element in addition yields

$$(q_b + q_{r\mu} - q_a - q_j) \langle b | J_{r\mu j} | a \rangle = 0.$$

A nonzero transition therefore has equal total charge before and after. A complete record final state may contain an emitted pair or several transport quanta; $q_{r\mu}$ always denotes the total charge of that environment final state. If the actual readout merges several charge values, the sum over μ keeps their statistical mixture; a single definite charge difference cannot be assigned to this coarse outcome.

Multiplying Eq. (30) on the left by $J_{r\mu j}^\dagger$, summing, and using the completeness relation of all Kraus operators gives

$$\mathcal{E}^*(Q_S) - Q_S = \sum_{r, \mu, j} (q_j - q_{r\mu}) J_{r\mu j}^\dagger J_{r\mu j}. \quad (31)$$

When the environment has incident occupation, the change of the system is determined by the difference of incoming and outgoing charges. For vacuum incidence, if the vacuum charge is taken as zero, the total charge of the record final state $|r\rangle$ as q_r , and the no-record action as M , this reduces to

$$[Q_S, M] = 0, \quad \mathcal{E}^*(Q_S) = Q_S - \sum_{r \neq 0} q_r J_r^\dagger J_r.$$

Taking $Q_S = H$ and writing ω_r for the energy of the complete outgoing environment state gives

$$\mathcal{E}^*(H) = H - \sum_{r \neq 0} \omega_r J_r^\dagger J_r.$$

The emitted pair of transit, the unretrieved quantum, and the transport products are all counted according to the actual final states, while the internal outflow and the subsequent retrieval are still composed coherently within the same no-record amplitude.

When conditioning on a single record, the incident statistics is also reweighted by that outcome. Let the outcome effect be $E_r^{\text{rec}} = \sum_{\mu, j} J_{r\mu j}^\dagger J_{r\mu j}$ and $w_r = \text{Tr}(E_r^{\text{rec}} \rho)$. Eq. (30) and its adjoint give $[E_r^{\text{rec}}, Q_S] = 0$; for $w_r > 0$,

$$\frac{\text{Tr}[Q_S \mathcal{I}_r(\rho)]}{w_r} = \frac{\text{Tr}[Q_S E_r^{\text{rec}} \rho]}{w_r} + \frac{\sum_{\mu, j} (q_j - q_{r\mu}) \text{Tr}[J_{r\mu j}^\dagger J_{r\mu j} \rho]}{w_r}.$$

The first term on the right is the incident mean weighted by that outcome; in general it differs from the unconditioned $\text{Tr}(Q_S \rho)$. The branch-by-branch balance of initial and final charges and the change of the conditional sample mean can therefore hold at the same time.

The balance of energy and momentum is carried in observational spacetime by event readings. We adopt the Hermitian energy-momentum reading of reference [4], written here as R_{ev} ; events satisfying the corresponding exchange matching give

$$\sum_{\text{out}} R_{\text{ev}} = \sum_{\text{in}} R_{\text{ev}}.$$

Its trace and traceless components give the balance of energy and momentum respectively. The angular-momentum reading combines the orbital and orientation contributions according to the existing readout; writing $\mathbf{J}_{\text{ev}}$ for this total angular-momentum reading, in a realization with joint rotational conservation one has

$$\sum_{\text{out}} \mathbf{J}_{\text{ev}} = \sum_{\text{in}} \mathbf{J}_{\text{ev}}.$$

The branch statement of a definite angular-momentum component is used for the readout along the corresponding axis; the joint rotation relation does not require the non-commuting components to take definite values simultaneously in the same record. Electric charge is balanced by counting all charged final states, and character charges of finite groups are balanced by the group multiplication of Eq. (28).

The normalization of the probabilities of all outcomes comes from joint unitarity and complete readout (§3.2). In the vacuum decomposition,

$$\mathcal{E}^*(\mathbf{1}) = \mathbf{1}, \quad M^\dagger M + \sum_{r \neq 0} J_r^\dagger J_r = \mathbf{1}.$$

The share of a single branch may decrease, while the sum of the no-record share at the end of the cycle and all record shares stays one. The balance of joint totals and the normalization of outcome shares jointly constrain the record statistics; the occupation of each sector of the system must still be judged from all branches.

5.3 Occupation and Equilibrium

For a sector of the system to keep its occupation in the cycle, no outcome may send states out of or into that sector. When the environment carries no corresponding charge, this condition can be given by the intertwining relation of the charge; in general the complete system map must be checked directly.

Let P be the orthogonal projection onto the sector under consideration, and for the moment let $\{J_\ell\}$ be all Kraus operators, including that of the no-record outcome. For the finite realization of §2,

$$\boxed{\mathcal{E}^*(P) = P \iff [J_\ell, P] = 0 \text{ for every } \ell.} \quad (32)$$

If every branch commutes with P , the completeness relation directly gives

$$\mathcal{E}^*(P) = \sum_{\ell} J_\ell^\dagger P J_\ell = P \sum_{\ell} J_\ell^\dagger J_\ell = P.$$

Conversely, if $\mathcal{E}^*(P) = P$, then for $u \in \text{Ran } P$ and $v \in \ker P$,

$$\begin{aligned} \sum_{\ell} \|(1 - P)J_\ell u\|^2 &= \langle u | 1 - \mathcal{E}^*(P) | u \rangle = 0, \\ \sum_{\ell} \|P J_\ell v\|^2 &= \langle v | \mathcal{E}^*(P) | v \rangle = 0. \end{aligned}$$

Every term is nonnegative, so each J_ℓ preserves both the range and the kernel of P , which gives the required commutation relations. Then $\text{Tr}[P\mathcal{E}(\rho)] = \text{Tr}(P\rho)$ holds for every incident state. Checking $[M, P] = 0$ alone is not sufficient for occupation preservation in the complete cycle. The preservation of the mean of a single self-adjoint quantity for all incident states, $\mathcal{E}^*(Q_S) = Q_S$, is in general still not sufficient for the preservation of all its spectral projections.

If the complete map moreover has a faithful stationary state $\sigma > 0$ with $\mathcal{E}(\sigma) = \sigma$, then mean preservation further implies the preservation of all spectral sectors. Using $Q_S = Q_S^\dagger$, $\mathcal{E}^*(\mathbf{1}) = \mathbf{1}$, and $\mathcal{E}^*(Q_S) = Q_S$,

$$\begin{aligned} \sum_{\ell} [Q_S, J_{\ell}]^{\dagger} [Q_S, J_{\ell}] &= \mathcal{E}^*(Q_S^2) - \mathcal{E}^*(Q_S)Q_S - Q_S\mathcal{E}^*(Q_S) + Q_S^2 \\ &= \mathcal{E}^*(Q_S^2) - Q_S^2. \end{aligned}$$

The expectation of the right side in σ is

$$\text{Tr}[\sigma(\mathcal{E}^*(Q_S^2) - Q_S^2)] = \text{Tr}[(\mathcal{E}(\sigma) - \sigma)Q_S^2] = 0.$$

Each term on the left is a positive semidefinite operator and σ is faithful, so $[Q_S, J_{\ell}] = 0$ for all branches. The finite-dimensional spectral decomposition then shows that every spectral projection of Q_S commutes with the J_{ℓ} , and Eq. (32) gives their preservation. Uniqueness of the stationary state is not required here. For a continuous-time quantum dynamical semigroup with a faithful stationary state, the fixed points in the Heisenberg picture form an operator algebra[22]; the argument above applies to the finite discrete maps of this paper.

In the vacuum restriction, if $W\psi = U(\psi \otimes |\text{vac}\rangle)$ satisfies

$$(Q_S \otimes \mathbf{1})W = WQ_S,$$

then taking matrix elements term by term in the environment basis shows that every branch commutes with Q_S , so all of its spectral sectors satisfy Eq. (32). If the environment carries a charge difference, Eq. (30) allows the system to cross the corresponding charge sectors, and joint conservation is still expressed by Eq. (29).

The preservation of shell occupation holds within the restriction to evaluation actions. The shell-preserving evaluation algebra of reference [3] and its inner fluctuations commute with the shell projections; when its self-adjoint actions and jump operators are adopted, each shell occupation is preserved. The conditions used here are shell-wise definition, avoidance of the poles of spectral evaluation, and preservation of shell blocks by finite Dirac-type actions; complex conjugation is treated as a real structure and conditional expectation as a map on operators. Once the cross-shell transport carried by $V, V^\dagger$ is added, probability currents appear between shells and the shell occupation begins to redistribute. The preservation relation after composing general transit and transport must be verified for all branches by Eq. (32); the cycle-by-cycle refreshing of the environment does not by itself provide this commutativity.

The preservation of the generation label in thermal transport is further limited by the tower depth and the time scale. In a finite tower the label can mix through the zero element; the deep-shell suppression makes it an effective superselection quantity within a certain observation time, with the leak rate and range of validity given in §4.5[3]. This change and the quark transit link of §5.1 are computed separately; the two processes cannot be rewritten together as an unconditional conservation of the generation charge. The character moments of the position distribution and the conserved charges are also defined separately: the unit-scaling symmetry of the position kernel allows the corresponding moments to decay with thermalization, and does not mean that every record branch preserves the system character sector[3].

Equilibrium in an environment at the same temperature is determined by joint energy conservation. In the chosen finite energy model, $\sigma_{\beta} \otimes \omega_{\beta}$ is the Gibbs weight of the total energy, from which §2 has given $\mathcal{E}_{\beta}(\sigma_{\beta}) = \sigma_{\beta}$ in Eq. (9). For a finite inverse temperature σ_{β} is faithful, so the self-adjoint quantities whose means are preserved under this map also satisfy the spectral-sector preservation conclusion above.

With

$$S(\rho \parallel \sigma) = \text{Tr}[\rho(\log \rho - \log \sigma)]$$

denoting the relative entropy, the data-processing inequality for completely positive trace-preserving maps[20, 21] gives

$$S(\mathcal{E}_\beta(\rho) \parallel \sigma_\beta) = S(\mathcal{E}_\beta(\rho) \parallel \mathcal{E}_\beta(\sigma_\beta)) \leq S(\rho \parallel \sigma_\beta).$$

Environment refreshing under the same incident conditions allows the same map to be iterated, so the relative entropy is nonincreasing cycle after cycle. When a compatible continuous limit forms a quantum dynamical semigroup with σ_β as stationary state, the corresponding relative-entropy dissipation rate is nonnegative[23]. Uniqueness of the equilibrium and convergence to it still depend on the connectivity of the actual channels and on the coherence they retain; the identification of the shell-occupation equilibrium, the position-level distribution, and the full-space Gibbs state follows the model conditions of §2. The conservation of joint totals, the preservation of system sectors, and the monotone approach to equilibrium thus each have a definite range of application.

6 Phenomena of the Dynamics

After its formation a standing wave propagates cycle by cycle, exchanges with the environment, and leaves distinguishable outcomes in the records. Identity-preserving exchanges change its state of motion, admissible transits terminate the original identity, and the reverse exchange can replenish the occupation of that identity. The evolution of a single particle and the change of occupation of many particles are both determined by the joint amplitude within the cycle and the record statistics after the cycle.

6.1 Quantum Motion

A stable particle can remain coherent along its propagation and leave discrete impact points at the place of detection; when the record conditions along the way are changed, the interference fringes change accordingly[4]. The propagation of the particle, the formation of the impact point, and the visibility of the fringes must therefore each be determined within the same cycle.

Without transit and transport the outflow component is completely retrieved within the cycle, and the single-cycle transfer follows $U_0 = CS$ of §2.1. In the position representation, write the corresponding action as

$$U_{\text{pos}} = \mathcal{F}^{-1}U_0\mathcal{F}.$$

Its propagation phase, dispersion relation, and wave-packet motion are given by §2.2 and reference [4]. Keeping this propagation action, we compute how environment records change the conditional state and the interference statistics.

A position record must provide spatial resolution and satisfy the energy balance of the exchange at the same time. For a non-flat dispersion of motion, an ideal record action that preserves every position eigenstate completely is limited by the conservation of total energy. Appendix A.2 constructs, under a specified coherent detection preparation, a finite joint exchange that preserves the total energy and realizes the target record action below with controlled error; the energy spectrum, couplings, and preparation of the reference are given as inputs.

Fix the particle identity and one propagation branch, and let ρ describe the position degree of freedom in this restriction. The lattice ring has N_x positions with coordinates $x_\ell = \ell/v$; the dual modular momentum transfers are $\Delta_q = 2\pi vq/N_x$, with $\ell, q = 0, \dots, N_x - 1$. Let X_{pos} be the

position operator. Write the total share of the target record as $0 < p_{\text{rec}} \leq 1$ and the amplitude of each momentum transfer as a_q , satisfying

$$\sum_q |a_q|^2 = p_{\text{rec}}, \quad J_q = a_q e^{-i\Delta_q X_{\text{pos}}}.$$

Denote the record pointer by D , with initial state $|d_{\text{in}}\rangle_D$. The isometric action of the target exchange is defined as

$$W_{\text{id}}|\psi\rangle = \sqrt{1 - p_{\text{rec}}} |\psi\rangle|\emptyset\rangle + \sum_q J_q |\psi\rangle|q\rangle_D. \quad (33)$$

The pointer states $|q\rangle_D$ are mutually orthogonal, and the outgoing no-response state $|\emptyset\rangle$ is orthogonal to all of them. Since $\sum_q J_q^\dagger J_q = p_{\text{rec}} \mathbf{1}$, the right side preserves inner products. This isometry secures the existence of some joint unitary extension; the balance with the given energy operator is realized by the reference construction in the appendix, whose record action approaches Eq. (33) under a finite reference.

If the detector resolves q directly, the normalized system action is the momentum transfer $\rho \mapsto e^{-i\Delta_q X_{\text{pos}}} \rho e^{i\Delta_q X_{\text{pos}}}$, and the position probability is unchanged. Resolving position requires the detector to combine these outgoing modes coherently. Let its readout basis be

$$|y\rangle_D = \frac{1}{\sqrt{N_x}} \sum_q e^{-i\Delta_q y} |q\rangle_D,$$

with y running over the same lattice ring. Define the normalized amplitudes $b_q = a_q / \sqrt{p_{\text{rec}}}$ and the detection response amplitude

$$h(d) = \frac{1}{\sqrt{N_x}} \sum_q b_q e^{i\Delta_q d}.$$

Fourier orthogonality gives $\sum_y |h(y - x)|^2 = 1$. The action of the impact point y is therefore

$$L_y = \frac{1}{\sqrt{N_x}} \sum_q e^{i\Delta_q y} J_q = \sqrt{p_{\text{rec}}} h(y - X_{\text{pos}}), \quad \sum_y L_y^\dagger L_y = p_{\text{rec}} \mathbf{1}. \quad (34)$$

For the state ρ before the exchange, the outcome share and the conditional state at the exchange are

$$p_y = p_{\text{rec}} \sum_x |h(y - x)|^2 \rho_{xx},$$

$$(\rho_y)_{xx'} = \frac{h(y - x) \overline{h(y - x')} \rho_{xx'}}{\sum_z |h(y - z)|^2 \rho_{zz}}, \quad p_y > 0.$$

The outcome y weights the original position distribution by $|h(y - x)|^2$; when the response is concentrated near zero, this weighting resolves positions near y . In particular, if all $b_q = 1/\sqrt{N_x}$, then

$$h(y - x) = \delta_{yx}, \quad p_y = p_{\text{rec}} \rho_{yy}, \quad \rho_y = |y\rangle\langle y|.$$

This gives the ideal position resolution within the target action. In the actual order of the cycle, the complete record operator is $U_{\text{pos}} L_y$ and the conditional state at the end of the cycle is $U_{\text{pos}} \rho_y U_{\text{pos}}^\dagger$; propagation continues after the record.

An actual position readout requires the coherent combination listed above and the energy exchange of the reference to be realized together. If unread energies or incident labels retain distinguishable information, the corresponding Kraus terms must be counted separately; merging classical

labels already formed does not produce conditional localization. The coherent reference of the appendix is in general not in a thermal state, and an environment at the same temperature does not by itself provide this detection preparation.

Denote the post-exchange system maps of the target and of the finite reference realization by $\mathcal{E}_{\text{id}}$ and $\mathcal{E}_L$, with L the range of coherent preparation of the reference over the level indices. The single-action error given in the appendix satisfies $\delta_L \leq 2\sqrt{N_x/L}$. For any initial state, the system trace distance and the complete outcome distribution satisfy

$$\frac{1}{2}\|\mathcal{E}_L(\rho) - \mathcal{E}_{\text{id}}(\rho)\|_1 \leq \delta_L, \quad \frac{1}{2}\sum_r |p_r^{(L)} - p_r^{\text{id}}| \leq \delta_L. \quad (35)$$

Here r includes the no-response outcome, and $p_y^{\text{id}} = p_y$. A subsequent identical U_{pos} does not change these bounds. The error of a normalized conditional state contains in addition the denominator of the outcome share, so outcomes with share near zero must be controlled separately.

Ignoring the impact point, the Fourier readout and the momentum-transfer readout give the same non-selective action:

$$\sum_y L_y \rho L_y^\dagger = \sum_q J_q \rho J_q^\dagger.$$

Including the no-response part, the post-exchange density matrix $\rho' = \mathcal{E}_{\text{id}}(\rho)$ given by the target action is

$$\rho'_{xx'} = g(x - x')\rho_{xx'}, \quad g(d) = 1 - p_{\text{rec}} + \sum_q |a_q|^2 e^{-i\Delta_q d}. \quad (36)$$

Environmental scattering can suppress the coherence terms in the position representation[24, 25]. In the target action above, $g(0) = 1$ keeps the position-diagonal elements; the off-diagonal elements are determined by the overlap of the pointer final states corresponding to different positions. This is the restriction of the overlap matrix of §3.1 to the position branch. Localization conditional on a specific impact point and decoherence after ignoring the outcome are given by different readouts of the same target output. The deviation of the actual finite exchange from this decoherence action is controlled by Eq. (35), and the bound does not require the position-diagonal elements of the finite exchange to be preserved term by term.

The visibility of two-path interference is connected with the path information retained in the environment[26]. For two paths leading to the same detection position, write the amplitudes as A_1, A_2 and the environment final states as $|e_1\rangle, |e_2\rangle$. The overlap formula of §3.1 gives

$$I = |A_1|^2 + |A_2|^2 + 2\text{Re}[A_1 \overline{A_2} \langle e_2 | e_1 \rangle].$$

Fixing the two path intensities and scanning the relative phase, and defining the fringe visibility by $\mathcal{V}_{\text{fr}} = (I_{\text{max}} - I_{\text{min}})/(I_{\text{max}} + I_{\text{min}})$, we obtain

$$\mathcal{V}_{\text{fr}} = \mathcal{V}_0 |\langle e_2 | e_1 \rangle|, \quad \mathcal{V}_0 = \frac{2|A_1||A_2|}{|A_1|^2 + |A_2|^2}.$$

In the target position-record model above, when the position difference of the two paths at the record is fixed at $d = x_1 - x_2$, this overlap is exactly $g(d)$. If every target action preserves the two path components and their separation, and an independent fresh environment is used each time, the coherence factor after n actions is $g(d)^n$; for general paths of motion the overlap must be computed from the actual positions and environment states of each action. When the actual cycle uses a freshly prepared copy of the same finite reference in every cycle, the accumulated trace-distance error of n actions does not exceed $n\delta_L$; this estimate applies within the time range in which that error remains controlled.

In the ideal limit of a dichotomous record, take both path amplitudes nonzero, suppose that a share f_{rec} of the environment outcomes resolves the two paths completely while the remainder keeps the original coherence, and that this share does not depend on the path. When the two parts are composed as orthogonal record branches, the environment overlap is $1 - f_{\text{rec}}$, and hence

$$\frac{\mathcal{V}_{\text{fr}}}{\mathcal{V}_0} = 1 - f_{\text{rec}}. \quad (37)$$

The relation of reference [4] between the completed share of the record and the lost share of the fringes holds in this dichotomous-record limit. For general partial overlap the complete $\langle e_2 | e_1 \rangle$ is used; the record count alone is not sufficient to determine the fringe visibility.

The actual visibility further requires control of the total share of the detection signal. For one position-record action, comparing fringes with the same incident normalization, let $s_{\text{id}} = I_{\text{max}}^{\text{id}} + I_{\text{min}}^{\text{id}}$. For $s_{\text{id}} > 2\delta_L$ the appendix gives

$$|\mathcal{V}_{\text{fr}}^{(L)} - \mathcal{V}_{\text{fr}}^{\text{id}}| \leq \frac{2\delta_L}{s_{\text{id}} - 2\delta_L}.$$

Here $\mathcal{V}_{\text{fr}}^{\text{id}}$ is the visibility of the target model above; Eq. (37) applies in its dichotomous-record limit. If n independently prepared actions are included, δ_L is replaced by $n\delta_L$ in the bound, with the requirement $s_{\text{id}} > 2n\delta_L$.

The propagation of a stable particle before and after a record is carried by the same single-cycle transfer. The impact point depends on the spatial resolution of the detection, and the fringes on the environment overlap left by the different paths; identity-preserving records can change both readouts. The records of an unstable particle can also mark the termination of its original identity.

6.2 Thermalization and Decay

An unstable particle in a medium can change its momentum through exchange and can also terminate its original identity through an admissible transit[5, 3]. Within one identity history, the momentum relaxation, the identity-preserving records, and the first exit are determined by the corresponding branches of the complete cycle; the specific decay process is identified by the physical final state of the exit record.

Take the finite realization of Appendix A.4. Two eigenmodes a_0, a_1 belong to the same species of particle, with different momenta and energies $E_0 < E_1$; they use the resonance ratio of the same exchange station and the same transit angle, and take part in same-temperature transport after retrieval. Let P be the projection onto the identity sector formed by these two modes, and ε the total share per cycle of exiting this sector. The fixed retrieval angle of §2 makes the exit contain two terms, the emitted pair and the unretrieved single quantum; the latter carries the energy and charge of the outflow, and its actual environment carrier must still be matched. The incident preparation label of the transit environment is kept, so that newly produced exit records are distinguishable from the existing thermal excitations. Keeping all non-exit outcomes, including identity-preserving transport records, we obtain

$$\mathcal{Q} = (1 - \varepsilon)\hat{\mathcal{Q}}, \quad \hat{\mathcal{Q}} = \text{Ad}_{U_0} \circ \mathcal{T}_\beta.$$

Here $\mathcal{T}_\beta$ is the reduced map of the transport segment and $\text{Ad}_{U_0}(\rho) = U_0 \rho U_0^\dagger$. The conditional state evolves by $\hat{\mathcal{Q}}$; $\mathcal{Q}$ evolves the unnormalized state, whose trace gives the identity-maintenance probability.

Write $\Delta E = E_1 - E_0$; the same-temperature occupation is

$$q_g = \frac{1}{1 + e^{-\beta\Delta E}}, \quad q_e = \frac{e^{-\beta\Delta E}}{1 + e^{-\beta\Delta E}}, \quad \sigma_P = q_g|a_0\rangle\langle a_0| + q_e|a_1\rangle\langle a_1|.$$

Let the transport angle be ξ , set $c_\xi = \cos \xi$, and write φ_i for the corresponding cycle eigenphases. The four Kraus terms given in the appendix map the space of all 2×2 matrices into itself; for any initial state ρ_0 their iteration satisfies

$$\begin{aligned} (\hat{\mathcal{Q}}^n \rho_0)_{00} &= q_g + c_\xi^{2n} [(\rho_0)_{00} - q_g], \\ (\hat{\mathcal{Q}}^n \rho_0)_{01} &= [c_\xi e^{i(\varphi_0 - \varphi_1)}]^n (\rho_0)_{01}. \end{aligned}$$

For $0 < \xi < \pi/2$ both the occupation deviation and the coherence component tend to zero, and the conditional state tends to σ_P . The decay rates of occupation and coherence are $-2v \log c_\xi$ and $-v \log c_\xi$ respectively; the relaxation of the complete conditional state must include both.

The same iteration also gives

$$\mathcal{Q}^n(\rho_0) = (1 - \varepsilon)^n \hat{\mathcal{Q}}^n(\rho_0), \quad P_P(n) = \text{Tr } \mathcal{Q}^n(\rho_0) = (1 - \varepsilon)^n.$$

The cycle number of the first exit is N_P , with waiting time $T_P = N_P/v$. From the difference of the identity-maintenance probabilities of adjacent cycles,

$$\Pr(N_P = n) = \varepsilon(1 - \varepsilon)^{n-1}, \quad \mathbb{E}T_P = \frac{1}{v\varepsilon}.$$

This waiting time refers to all exit records; the actual weak-decay lifetime must be matched with the terminating record group determined by the corresponding physical final state.

Let $D(\rho, \sigma) = \frac{1}{2}\|\rho - \sigma\|_1$ be the trace distance. The complete conditional spectrum gives

$$D(\hat{\mathcal{Q}}^n \rho_0, \sigma_P) \leq c_\xi^n D(\rho_0, \sigma_P).$$

For an error tolerance $0 < \delta_{\text{tol}} < 1$, take

$$n_{\text{mix}}(\delta_{\text{tol}}) = \max \left\{ 1, \left\lceil \frac{\log(1/\delta_{\text{tol}})}{-\log c_\xi} \right\rceil \right\}.$$

When $n_{\text{mix}}[-\log(1 - \varepsilon)] \ll 1$, the original particle still keeps its identity with probability close to one by the time the conditional state is close to equilibrium. Thermal transport can therefore take place within one identity history without changing the geometric exit law of this realization.

The stability of an identity must be judged against all branches actually taking part. When the identity-changing channels are closed, the criterion $\mathcal{E}^*(P) = P$ of §5.3 is equivalent to all Kraus operators commuting with P , which secures that every record preserves that identity. With a nonzero thermal label leak, strict preservation becomes effective preservation within a finite time; if the leak rate is Γ_{sel} , then $t_{\text{obs}}\Gamma_{\text{sel}} \ll 1$ delimits the corresponding superselection time window. The deep-shell suppression given in reference [3] determines this window under its energy and transport matching conditions and cannot replace the branch check of a general cycle.

Identity-preserving records can also complete the readout of the incident direction. To distinguish it from momentum, label the character directions of §3.3 by s , and let P_s be their orthogonal projections in the chosen propagation space. Require $[H, P_s] = [U_0, P_s] = 0$, and require P_s to commute with the system momentum and the conserved charges taking part in the present balance; each sector may contain several momentum modes. The detection set contains all non-principal

directions with nonzero weight in the preparation used. Let the number of selected directions be d , and add a finite detection environment: the vacuum energy is zero, the carrier states $|d_0\rangle, |d_s\rangle$ have the same energy $E_d > 0$, the same total momentum, and the same verified conserved charges. The positive-temperature incident Gibbs state is

$$\omega_{\beta,d} = Z_d^{-1} |\text{vac}\rangle\langle\text{vac}| + w_d \left(|d_0\rangle\langle d_0| + \sum_s |d_s\rangle\langle d_s| \right),$$

$$Z_d = 1 + (d+1)e^{-\beta E_d}, \quad w_d = \frac{e^{-\beta E_d}}{Z_d}.$$

The environment modes, the degeneracy, the incident preparation, and the scattering couplings below are inputs of this detection realization. Since the carrier energy is positive, its incident occupation vanishes in the zero-temperature limit and this detector no longer responds; this limitation concerns the chosen preparation. The zero-frequency coupling allowed by §2.3 can be taken as

$$K_{\text{det}} = \sum_s \theta_s^{\text{det}} P_s \otimes (|d_s\rangle\langle d_0| + |d_0\rangle\langle d_s|), \quad 0 < \theta_s^{\text{det}} \leq \frac{\pi}{2},$$

whose action $Y_{\text{det}} = e^{-iK_{\text{det}}}$ is the identity on the unlisted complement. The different P_s are orthogonal, so each direction acts separately on the corresponding two-state environment block; equal energy, equal momentum, and equal charge secure the joint balance. This action does not send the system out of P_s , and the scattered carrier leaves with the actual record.

The incident label j is kept cycle by cycle, and the outgoing label r is read out. This example needs the incident label to be actually kept: every controlled rotation preserves $\omega_{\beta,d}$, so if one looks only at the outgoing environment, the reduced environment state is the same Gibbs state for every direction, and the incident direction cannot be distinguished from it. For direction s , the two outcomes $(j, r) = (d_0, d_s)$ and (d_s, d_0) each have one Kraus operator supported only on P_s ; including U_0 , both are

$$-i\sqrt{w_d} \sin \theta_s^{\text{det}} U_0 P_s.$$

Merging these two outcomes into the direction response $\mathcal{I}_s$ and writing the single-cycle response share as $q_{s,\text{det}}$, its effect is

$$\mathcal{I}_s^*(\mathbf{1}) = q_{s,\text{det}} P_s, \quad q_{s,\text{det}} = 2w_d \sin^2 \theta_s^{\text{det}} > 0.$$

All remaining outcomes go into the sub-map $\mathcal{N}$ of outcomes that have not yet acquired a direction label; it keeps the complete incident and outgoing branches, unlike the single no-record amplitude of the vacuum restriction in §3.3. The completeness relation and sector preservation give

$$\mathcal{N}^*(P_s) = (1 - q_{s,\text{det}}) P_s.$$

The effect of acquiring no direction label in the first $n-1$ cycles and reading out s in the n -th is therefore $(\mathcal{N}^*)^{n-1}(q_{s,\text{det}} P_s)$. Summing over all possible first-response times,

$$Q_{s,N} = \sum_{n=1}^N (\mathcal{N}^*)^{n-1}(q_{s,\text{det}} P_s) = [1 - (1 - q_{s,\text{det}})^N] P_s \longrightarrow P_s. \quad (38)$$

This realizes direction preservation, record resolution, and the completeness of repeated readout simultaneously at positive temperature. If an intra-identity transport satisfying $\mathcal{G}^*(P_s) = P_s$ is composed first, then

$$(\mathcal{I}_s \circ \mathcal{G})^*(\mathbf{1}) = \mathcal{G}^*(q_{s,\text{det}} P_s) = q_{s,\text{det}} P_s, \quad (\mathcal{N} \circ \mathcal{G})^*(P_s) = (1 - q_{s,\text{det}}) P_s.$$

The two effect relations therefore still hold, without assuming that this transport commutes with the detection action. The two momentum modes of Appendix A.4 belong to the same P_s ; the detection environment is independent of the two existing parts of the environment and is composed as an equal-energy exchange factor of Y before the existing U_0 . The non-selective action of the scattering Y_{det} on the system within a single P_s is the identity, so it does not change the thermalization formulas above inside that identity.

The Born distribution after the direction response follows the preparation of §3.3. The isometric embedding sends $|\chi_s\rangle$ to the eigenmode $|u_s\rangle$ in the corresponding sector; the surviving preparation inside formation exchange gives the non-principal direction amplitudes $R_s/\sqrt{p-1}$. Here p is the fixed odd prime and R_s the resonance ratio of reference [2]. By the effect above, after waiting N cycles the distribution conditional on detecting a non-principal direction is

$$p_s^{(N),\text{cond}} = \frac{|R_s|^2[1 - (1 - q_{s,\text{det}})^N]}{\sum_{t \neq 0} |R_t|^2[1 - (1 - q_{t,\text{det}})^N]} \longrightarrow \frac{|R_s|^2}{\sum_{t \neq 0} |R_t|^2}. \quad (39)$$

The conditional distribution is defined only when the denominator is nonzero. The embedding, the coherent preparation, and the statistical conditions of §3.3 remain in force; this detection realization gives an actual repeated readout of stable directions.

For an unstable particle, the first exit and the direction response also compete. In a realization with exit before detection, if the non-exit action preserves P_s with a constant exit share ε_s for every initial state in it, and only identity-preserving detection responses are accepted, then the accepted share per cycle is $(1 - \varepsilon_s)q_{s,\text{det}}$ and the share that continues to wait is $(1 - \varepsilon_s)(1 - q_{s,\text{det}})$. The final detection efficiency is

$$\eta_s^{\text{det}} = \frac{(1 - \varepsilon_s)q_{s,\text{det}}}{\varepsilon_s + (1 - \varepsilon_s)q_{s,\text{det}}}. \quad (40)$$

When the efficiencies of different directions differ, the conditional distribution keeps this weighting. If the exit records can also resolve the incident direction, the corresponding complete readout is recovered only by including them in their respective terminating record groups; the stable restriction $\varepsilon_s = 0$ directly gives $\eta_s^{\text{det}} = 1$.

A particle can thermalize and be identified while its identity is maintained; an identity history can also be terminated by transit after these processes. The corresponding shares and rates are determined separately by the record groups and their complete cycle actions. The overall occupation of many particles is determined jointly by the exit of the original identity and the formation of new particles.

6.3 Formation and Equilibrium

In the realization with the nonzero constant exit share above, the identity-maintenance probability of a single particle tends to zero, while an ensemble under continued supply can keep a nonzero occupation. The overall occupation depends on the exit of the original particles and the formation of new ones. Two kinds of background give different ways of replenishment: in a thermal background the reverse exchange is determined by the incident occupation, while an external drive continuously provides a specified formation input.

Forward transit and reverse formation are given by the same $W = F^\dagger T F$, with the level ordering $E_z, E_b < E_i$ of Appendix A.4. Let $c = |R_s|$, $X = 1 - c^2$, and let η be the transit angle. The squared moduli of the matrix elements of the two kinds of reverse formation are

$$d_{\text{pair}} = X \sin^2 \eta, \quad d_{\text{single}} = c^2 X (1 - \cos \eta)^2.$$

They return from $b \otimes \nu_i$ and $z \otimes 1_i$ to $a_i \otimes \text{vac}$. A zero-temperature vacuum without external energy supply does not occupy these incident modes, so the reverse replenishment of a_i vanishes; the decay of an existing higher-energy parent state in vacuum can still form lower-energy final states.

When system and environment are in complete same-temperature Gibbs equilibrium, let the vacuum weight of the transit environment be π_{vac} and the Gibbs partition function of the system Z_S . The weights of the two kinds of incident joint states are equal, both being

$$\frac{\pi_{\text{vac}}}{Z_S} e^{-\beta E_i}.$$

Hence the formation input of each momentum mode is

$$(A_m)_{ii} = \frac{\pi_{\text{vac}} e^{-\beta E_i}}{Z_S} (d_{\text{pair}} + d_{\text{single}}) = \varepsilon \frac{e^{-\beta E_i}}{Z_S}.$$

The subsequent transport and single-cycle transfer preserve the conditional Gibbs state σ_P of this identity, so the replenishment of the complete cycle satisfies

$$A_P = \varepsilon \pi_P \sigma_P, \quad \pi_P = \text{Tr}(P \sigma_\beta).$$

Adding the non-exit action gives

$$\pi_P \sigma_P = \pi_P \mathcal{Q}(\sigma_P) + A_P.$$

At equilibrium the total shares of formation and of exit per cycle are both $\varepsilon \pi_P$, and the expectations of both counts over N consecutive cycles are $N \varepsilon \pi_P$. The identity-maintenance probability of a batch of initial particles decreases as $(1 - \varepsilon)^N$, while the mean occupation in the system is replenished by reverse formation.

The local ratio of forward and reverse rates is directly determined by the paired matrix elements above and the Boltzmann weights. The Gibbs invariance of a general sequential cycle follows from joint energy conservation by §5.3; channel-by-channel detailed balance must be checked separately. To approach the Gibbs weights of the occupation by species from an arbitrary initial state, one must further verify the connectivity of the admissible channels and the decay of the corresponding nonstationary spectrum. When strictly preserved sectors exist, equilibrium also keeps the initial weight of each sector.

Under driven supply, take the linear restriction in which interactions between particles and occupation feedback are negligible. This description does not determine the exchange statistics of identical particles, nor does it include the formation and dissolution of bound composites or many-body condensation. Let $G_n \geq 0$ be the unnormalized occupation operator at the end of the cycle, and $N_n = \text{Tr } G_n$ the expected number of tracked particles. The initial occupation G_0 and the formation input Φ are both supported in P ; $\mathcal{Q}$ is a completely positive trace-non-increasing map preserving the support on this sector. If a fixed positive formation input Φ is added at the end of every cycle and the existing particles evolve by $\mathcal{Q}$, then

$$G_{n+1} = \mathcal{Q}(G_n) + \Phi. \tag{41}$$

Successive substitution gives

$$G_n = \mathcal{Q}^n(G_0) + \sum_{j=0}^{n-1} \mathcal{Q}^j(\Phi).$$

In a finite-dimensional space, if the spectral radius $r(\mathcal{Q}) < 1$, the first term tends to zero, the series converges, and the steady occupation is

$$G_* = (\mathbf{1} - \mathcal{Q})^{-1} \Phi = \sum_{j=0}^{\infty} \mathcal{Q}^j(\Phi). \quad (42)$$

Here $\mathbf{1}$ denotes the identity map on the operator space; every term is a positive semidefinite operator, so the steady occupation operator is positive semidefinite as well.

Let $f = \text{Tr } \Phi > 0$ and $\phi = \Phi/f$. The probability that a newly formed particle keeps its identity through the first j cycles is $\text{Tr } \mathcal{Q}^j(\phi)$. Using the tail-sum formula for the cycle number of the first exit,

$$\mathbb{E}_\phi T = \frac{1}{v} \sum_{j=0}^{\infty} \text{Tr } \mathcal{Q}^j(\phi).$$

Taking the trace of the steady-occupation series gives

$$N_* = v f \mathbb{E}_\phi T. \quad (43)$$

Here vf is the production rate and $\mathbb{E}_\phi T$ the mean first-exit waiting time computed for the actual formed state; when the exit records are exactly all decay records of the particle under consideration, it gives the corresponding mean lifetime. The mean occupation equals the input rate times the mean residence time, which is the statistical form of Little's law[27]; here the mean time is determined by the first-exit distribution of the chosen cycle and includes the internal evolution after formation. The time scale of exponential decay cannot stand in for this mean time.

When the exit share per cycle inside the identity sector is a constant $0 < \varepsilon < 1$, that is, $\mathcal{Q}^*(P) = (1 - \varepsilon)P$ on this sector, the trace of the occupation evolves autonomously:

$$N_{n+1} = (1 - \varepsilon)N_n + f, \quad N_n = (1 - \varepsilon)^n N_0 + \frac{f}{\varepsilon} [1 - (1 - \varepsilon)^n].$$

Hence $N_* = f/\varepsilon$. The first term belongs to the history of the original particles, the second to the particles that continue to be maintained after each replenishment. The linear formula with fixed Φ applies only when the supply reservoir is fixed and the cycle action of each particle does not change with the total occupation.

The normalized state of the steady occupation also depends on the formation input. If, following §6.2, $\mathcal{Q} = (1 - \varepsilon)\hat{\mathcal{Q}}$ and $\hat{\mathcal{Q}}(\sigma_P) = \sigma_P$, then $\Phi = f\sigma_P$ gives $G_* = (f/\varepsilon)\sigma_P$. Conversely, if the normalized steady occupation is exactly σ_P , substituting back into the steady-occupation equation yields $\Phi = f\sigma_P$. Therefore

$$\frac{G_*}{N_*} = \sigma_P \iff \frac{\Phi}{\text{Tr } \Phi} = \sigma_P \quad (44)$$

holds in this realization of constant exit and conditional Gibbs preservation.

The steady occupation under nonthermal supply can be computed directly. Let $q = 1 - \varepsilon$ and let the normalized steady occupation be $\rho_* = G_*/N_*$. On the two momentum modes of Appendix A.4, write $\lambda_{\text{coh}} = c_\xi e^{i(\varphi_0 - \varphi_1)}$; summing the steady-state series component by component gives

$$(\rho_*)_{00} = q_g + \frac{\varepsilon}{1 - qc_\xi^2}(\phi_{00} - q_g), \quad (\rho_*)_{01} = \frac{\varepsilon}{1 - q\lambda_{\text{coh}}} \phi_{01}.$$

The conditional state of a single batch of particles can tend to σ_P from a nonthermal initial state, whereas an ensemble under continued supply always contains more recently formed particles and

therefore in general keeps the deviation of the supply. Summing the trace-distance bound of §6.2 term by term,

$$D(\rho_*, \sigma_P) \leq \frac{\varepsilon}{1 - qc_\xi} D(\phi, \sigma_P).$$

When exit is much slower than the relaxation of the complete conditional state, that is, $\varepsilon \ll 1 - c_\xi$, a nonthermal supply can also produce an overall steady state close to the thermal distribution; its deviation is controlled by the bound above.

The common shell chain has a common normalized relaxation spectrum under the same conditions. The common shell chain of reference [3] contains no mass parameter of the particle species; for the same prime p , inverse temperature β , tower depth K , boundary, and couplings, its normalized nonstationary spectrum is

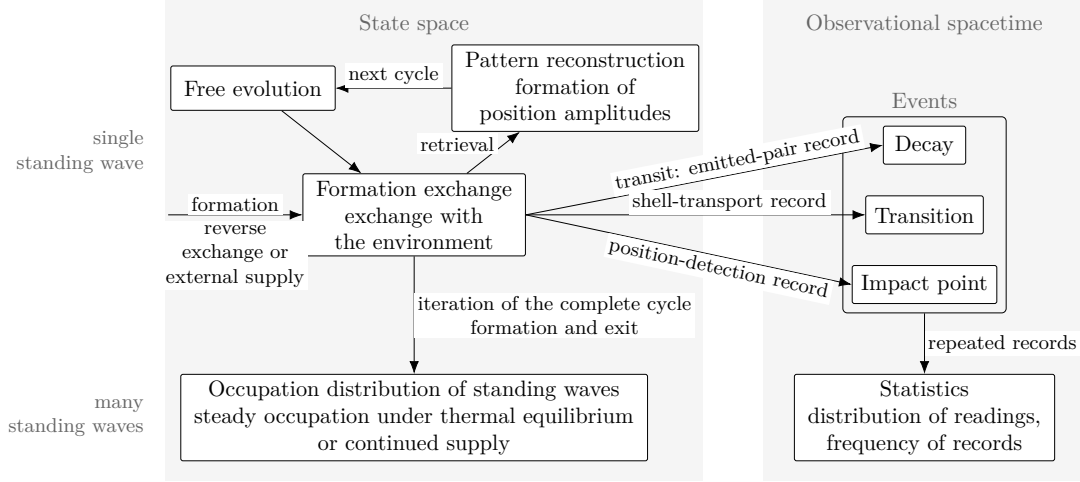
$$g_{\beta,K,r} = 1 - \frac{2p^{\beta/2}}{1 + p^\beta} \cos \frac{r\pi}{K+1}, \quad r = 1, \dots, K.$$

The actual rates of the continuous weak-collision description are $\kappa g_{\beta,K,r}$, with κ the total rate unit; a finite cycle uses the negative logarithm of the modulus of the corresponding eigenvalue as in §4.3. In the infinite-depth limit the normalized shell spectral gap depends only on p and β . Position transport, on the other hand, gives the equilibrium weights $e^{-\beta E_\alpha(k)}$ according to the dispersion of each particle, and its distribution and relaxation rates may still depend on the mass and on the matrix elements of the medium.

An environment at the same temperature fixes a common temperature, and a compatible shell structure gives a common normalized spectrum; formation and exit determine the occupation of each identity. In the linear-ensemble restriction the total occupation is obtained by adding the maintenance histories of the particles and the formation input, while the propagation, thermal transport, and terminating records within each history are determined by the same cycle.

7 Conclusion

Under the given exchange and environment conditions, this paper has composed free evolution, formation exchange, and pattern reconstruction into a unitary cycle on a finite joint space. The subsequent state of the system and the actual records in the outgoing environment are determined by the same joint evolution, and the system state at the end of the cycle constitutes the initial state of the next. Under the same incident preparation and cycle-by-cycle environment refreshing, the system evolves cycle after cycle by the same cycle action, and formation, motion, decay, and thermalization thereby obtain a common dynamical description.



The actions within the exchange and their connections to the processes are

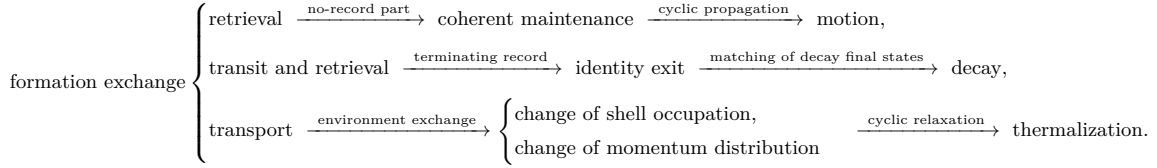


Figure 1: The common cycle of formation, motion, decay, and thermalization. The relations below describe the evolution of the system; the right column lists representative events read out from records. The terminating records of the transit block shown include the emitted pair and the unretrieved single quantum; actual decays must be identified by the corresponding physical final states. Records can preserve identity; position impact points use the specified coherent detection preparation and the error range listed in the text. The conditional state of a single standing wave can thermalize within the cycle, and the occupation of many standing waves is determined jointly by the individual histories. Formation requires the corresponding incident occupation or an external supply; thermalization, thermal equilibrium, and steady occupations use the conditions listed in the text.

After formation, a standing wave is maintained and moves within the cycle. Identity-preserving exchanges can produce outgoing records, and the target action of a position record is approached by a finite conserving exchange under the specified coherent preparation; intra-identity transport can moreover bring the conditional state to equilibrium when the relaxation conditions are met. When matched with the corresponding physical final state, a terminating record identifies the exit of an identity as a decay event. Motion, the formation of records, and conditional thermalization can therefore take place within one and the same history of identity maintenance, while decay

constitutes one way in which that history ends.

In the linear-ensemble description, the occupation of many standing waves is determined jointly by the identity-maintenance histories following each formation, and the accumulation of actual records gives the corresponding event statistics. The evolution of a single standing wave is thereby connected with the change of the overall occupation, and the processes in state space are connected through actual records with the events in observational spacetime. The common cycle constructed in this paper allows the relations among formation, motion, decay, and thermalization to be given by the same joint evolution and its iteration.

Data Availability

The main results of this paper are obtained by analytic derivation and use no experimental data. The auxiliary numerical verification scripts for the finite cycle, the record actions, the occupation spectra, and the supply model, together with their parameters and running instructions, are included in the numerical material accompanying the manuscript; all inputs serve to verify the finite models listed and are not used as experimental fits to actual particles or media.

Conflict of Interest

The author declares no conflict of interest.

A Finite Realization of the Common Cycle

The physical identification of the common cycle requires the section structure, the exchange energies, and the actual records to be compatible. The embedding of the section into the complete state space fixes the correspondence of the algebraic actions, and the energies and environment modes taking part in the exchange must in addition be balanced on the joint space.

A.1 Section and Zero Element

On the finite lattice space $L^2(\mathbb{Z}/p^K\mathbb{Z})$, a nonzero position is written $x = p^j \varpi(x)u(x)$, where j is the shell index, ϖ the Teichmüller unit, and u the principal-unit position. The normalized embedding of the section $L^2(\mathbb{Z}/p\mathbb{Z})$ into the tower space is defined as

$$\mathcal{J}_K \delta_c = p^{-(K-1)/2} \sum_{x \equiv c \pmod{p}} \delta_x.$$

The supports of different residue classes are disjoint and each class has p^{K-1} points, so $\mathcal{J}_K^\dagger \mathcal{J}_K = \mathbf{1}$. With the translation convention $(T_a f)(x) = f(x + a)$ of the main text, $T_a \delta_x = \delta_{x-a}$. Translation maps residue classes bijectively to residue classes, hence

$$T_a \mathcal{J}_K = \mathcal{J}_K T_{a \bmod p}.$$

Let P_Z be the projection onto the zero residue class $Z = \{x : p \mid x\}$ and $D_F = |\delta_0\rangle\langle\delta_0|$ the zero-element projection of the section; then

$$P_Z \mathcal{J}_K = \mathcal{J}_K D_F, \quad [P_Z, T_a] \mathcal{J}_K = \mathcal{J}_K [D_F, T_{a \bmod p}].$$

This proves the compatibility of the section vertex on the coarse position subspace. Multiplication by units also permutes the residue classes and is therefore compatible with the corresponding character projections.

The coarse zero vector $\mathcal{J}_K \delta_0$ differs from the single point $\delta_0^{(K)}$ of the tower; their overlap is $p^{-(K-1)/2}$. The share of the coarse zero vector on shell j , for $j = 1, \dots, K-1$, is $(p-1)/p^j$, and its share on the terminal point is $p^{-(K-1)}$; these shares sum to one. It therefore in general extends over several shells and cannot on this basis be regarded as a single eigenstate of the valuation energy.

The state $|z\rangle$ of §2 is a reference state in the effective energy description. Identifying it with the coarse zero vector further requires a common energy operator and a verification of the energy matching of the exchange; the intertwining relations of $\mathcal{J}_K$ above alone do not imply that H , the shell-raising shift, or the complete U preserves this subspace. In particular, in the description with shell energy $j \log p$, the coarse zero vector is not a zero-energy state, and whether the original rates remain unchanged after absorbing the projection factors of the shells into the retrieval angle must still be verified.

The shell-occupation spectrum follows the reflecting-boundary chain of §4.3 and the corresponding discrete transfer matching. Whether the algebraic relations given by the section embedding and these occupation results can enter the common cycle at the same time is still decided by the energies and the exchange matrix used.

A.2 Shell and Position Exchange

A position record is carried by the exchange of energy and momentum between the system and the medium. Fix the particle identity and one propagation branch, and take the finite lattice ring of reference [4], with positions $x_\ell = \ell/v$ and dual physical momenta $k_n = 2\pi v n/N$. The dispersion satisfies

$$\cos(E/v) = \cos(k/v) \cos(m_\alpha/v).$$

On the negative-phase propagation branch the single-cycle transfer is $U_{\text{pos}} = e^{-iE(k)/v}$; the complete doublet is described by CS . When the system transition $k \rightarrow k'$ is paired with an environment excitation, the joint matrix element of transport is

$$|k'\rangle\langle k| \otimes c_{(\Delta, \omega)}^\dagger + \text{h.c.}, \quad \Delta = k - k', \quad \omega = E(k) - E(k').$$

The environment excitation carries the momentum and energy lost by the system, so this matrix element commutes with the total momentum and with the total energy. The excitation spectrum of the medium and the corresponding couplings are inputs of this realization of transport.

When the chosen massive propagation branch satisfies $\sup_k |\partial E/\partial k| < 1$, the energy matching of free single-photon emission is restricted. A photon carries momentum $q \neq 0$ and energy $|q|$, whereas along the short arc of the finite momentum ring connecting the initial and final states, the mean-value theorem gives

$$|E(k) - E(k - q)| \leq \sup_k |\partial E/\partial k| |q| < |q|.$$

This free single-photon process therefore cannot satisfy energy conservation. If the incident environment is the vacuum and the position coupling contains only the free photon modes above, position transport produces no record. A medium can have a different excitation spectrum, and the corresponding exchange must still be matched according to the actual (Δ, ω) ; this exclusion does not apply to other carriers or to zero-frequency detection actions.

An energy-resolved isolated exchange block gives the record operator $J_{k'k} = -i \sin g |k'\rangle\langle k|$, where g is the dimensionless exchange angle of that block. The target momentum-transfer action

$J_\Delta = a_\Delta e^{-i\Delta X_{\text{pos}}}$, however, contains different energy differences. Decomposing it by the actual Bohr frequencies,

$$J_{\Delta,\omega} = a_\Delta \sum_{E(k)-E(k-\Delta)=\omega} |k-\Delta\rangle\langle k|, \quad [H, J_{\Delta,\omega}] = -\omega J_{\Delta,\omega}.$$

Here $k-\Delta$ is understood as an index on the finite ring, while the energy differences still use the corresponding eigenvalues. If the terms correspond to distinguishable complete environment final states, they must be kept separately in the reduction; merging classical labels cannot restore the interference between them. To obtain the coherent action of the whole J_Δ , a detection preparation capable of balancing these energy differences must be specified.

This requirement follows directly from exact conservation. Let the initial environment be a fixed pure state $|d\rangle$ uncorrelated with the system, and count all degrees of freedom that are not read out as part of the environment. Suppose the joint unitary satisfies $[U, H \otimes \mathbf{1} + \mathbf{1} \otimes H_B] = 0$ and that for every position eigenstate $U(|x\rangle|d\rangle) = |x\rangle|e_x\rangle$. Writing its isometric action as $W|\psi\rangle = U(|\psi\rangle|d\rangle)$, conservation gives

$$W^\dagger(H \otimes \mathbf{1} + \mathbf{1} \otimes H_B)W = H + \langle d|H_B|d\rangle\mathbf{1}.$$

Taking matrix elements with $x \neq x'$, the environment energy term vanishes because $\langle x|x'\rangle = 0$, hence

$$\boxed{H_{xx'}(1 - \langle e_x|e_{x'}\rangle) = 0 \quad (x \neq x').} \quad (45)$$

A mixed environment can be purified by adding an auxiliary space of zero energy, and the same conclusion holds. For the ideal position resolution with equal-amplitude a_q in §6.1, $\langle e_x|e_{x'}\rangle = 1 - p_{\text{rec}} + p_{\text{rec}}\delta_{xx'}$, so $p_{\text{rec}}H_{xx'} = 0$ is required. For a non-flat dispersion H has nonzero off-diagonal elements in the position representation, so for $p_{\text{rec}} > 0$ this target of completely preserving the position eigenstates cannot be realized exactly by such a conserving action. This is the form taken in the present model by the restriction that additive conservation imposes on repeatable measurements[28, 29]; it limits the target action above and does not exclude general position detection.

A finite coherent reference allows a conserving action to approach this target. The preparation of the coherent reference is given as a condition of the construction, and its role may be compared with reference [30]. Let $d = N_x$, let the common eigenbasis of the system energy and momentum be $\{|k_j\rangle\}_{j=1}^d$ with energies E_j , and take the energy and momentum of the record pointer D to be zero. On position states, Eq. (33) is written $W_{\text{id}}|x\rangle = |x\rangle|e_x\rangle$ with $|e_x\rangle$ normalized. For each x , extending $|d_{\text{in}}\rangle \mapsto |e_x\rangle$ to a unitary matrix U_x on the pointer gives

$$U_{\text{id}} = \sum_x |x\rangle\langle x| \otimes U_x = \sum_{i,j} |k_i\rangle\langle k_j| \otimes U_{\text{id},ij}.$$

Only the completion of a finite-dimensional orthonormal basis is used here; U_{id} itself is not required to commute with H .

Take an integer $L \geq 1$, and let the basis of the detection reference R be $|\mathbf{n}\rangle$ with $\mathbf{n} \in \{0, \dots, L+1\}^d$. Write $E_{\text{min}} = \min_j E_j$ and $\Delta E_j = E_j - E_{\text{min}}$, and specify

$$H_R|\mathbf{n}\rangle = \left(\sum_j \Delta E_j n_j \right) |\mathbf{n}\rangle, \quad |r_L\rangle = L^{-d/2} \sum_{\mathbf{n} \in \{1, \dots, L\}^d} |\mathbf{n}\rangle.$$

Let $\mathbf{e}_j$ be the j -th integer unit vector. Each system-and-reference basis vector $|k_j\rangle|\mathbf{n}\rangle$ belongs uniquely to the subspace labeled by $\mathbf{m} = \mathbf{n} + \mathbf{e}_j$. When all $1 \leq m_\ell \leq L+1$, this subspace contains

every j ; on it define

$$U_L(|k_j\rangle|\chi\rangle_D|\mathbf{m} - \mathbf{e}_j\rangle_R) = \sum_i |k_i\rangle U_{\text{id},ij}|\chi\rangle_D \otimes |\mathbf{m} - \mathbf{e}_i\rangle_R. \quad (46)$$

On the remaining subspaces set $U_L = \mathbf{1}$. The matrix on a complete subspace is the same as that of U_{id} , and the action on the remaining subspaces is the identity, so U_L is unitary on the whole finite space. The total energy of a complete subspace is $E_{\text{min}} + \sum_j \Delta E_j m_j$, independent of j , hence

$$[U_L, H \otimes \mathbf{1} + \mathbf{1} \otimes H_R] = 0.$$

The identity factor on the pointer is omitted here. Every basis vector of the initial reference state lies in one of the complete subspaces above, and the identity extension on the boundary does not affect the action on this initial state.

The same construction can also balance an ordinary additive momentum. Choosing any reference value k_{ref} , specify $\hat{k}_R|\mathbf{n}\rangle = \sum_j (k_j - k_{\text{ref}})n_j|\mathbf{n}\rangle$; the total momentum of a complete subspace is $k_{\text{ref}} + \sum_j (k_j - k_{\text{ref}})m_j$, again independent of j . Hence U_L commutes with the total momentum, including the actual momentum differences when indices wrap around. The system is always restricted to the same identity sector, and the identity extension on the remaining sectors preserves that identity. The energy spectrum and momentum spectrum of the reference, the degeneracy of the pointer, and the exchange couplings are all specified conditions of this realization.

Taking the Hermitian logarithm of the finite unitary matrix on each complete subspace gives $U_L = e^{-iK_L}$, with K_L commuting with the total energy and total momentum above. When K_L is expanded in the matrix units of the system energy eigenbasis, each environment coefficient takes up the opposite energy difference, so K_L can be written as the sum of the nonzero-frequency and zero-frequency terms listed in §2.3. This gives one finite realization of the transport segment. When transit is closed and the outflow component is completely retrieved, $F^\dagger T F = \mathbf{1}$ and the complete cycle can be taken as $(U_{\text{pos}} \otimes \mathbf{1})U_L$. Whether a specific medium provides this reference and coupling must still be checked against its physical modes; the dispersion relation itself does not determine these preparations.

The error is controlled by the overlap of the reference before and after translation. Define the normalized states in the finite space

$$|r_L^{ij}\rangle = L^{-d/2} \sum_{\mathbf{n} \in \{1, \dots, L\}^d} |\mathbf{n} + \mathbf{e}_j - \mathbf{e}_i\rangle.$$

For $i = j$ it equals $|r_L\rangle$; for $i \neq j$,

$$\langle r_L | r_L^{ij} \rangle = (1 - 1/L)^2, \quad \|r_L^{ij} - r_L\|^2 = 4/L - 2/L^2.$$

Only the required finite translated states are defined here; the translation on the bounded index range is not regarded as a unitary operator on the whole reference space. For any normalized input $|\Psi\rangle = \sum_j |k_j\rangle|\psi_j\rangle_D$, the difference between the actual output and $U_{\text{id}}|\Psi\rangle \otimes |r_L\rangle$ is

$$\sum_j \sum_i |k_i\rangle U_{\text{id},ij}|\psi_j\rangle \otimes (|r_L^{ij}\rangle - |r_L\rangle).$$

For fixed j the system components i are orthogonal, and $\sum_i U_{\text{id},ij}^\dagger U_{\text{id},ij} = \mathbf{1}$. The norm of this inner sum therefore does not exceed $\sqrt{4/L - 2/L^2} \|\psi_j\|$. Applying the triangle inequality and the

Cauchy–Schwarz inequality over j gives

$$\boxed{\begin{aligned} \|U_L(|\Psi\rangle|r_L\rangle) - (U_{\text{id}}|\Psi\rangle)|r_L\rangle\| &\leq \delta_L, \\ \delta_L &= \sqrt{d(4/L - 2/L^2)} \leq 2\sqrt{d/L}. \end{aligned}} \quad (47)$$

The same estimate allows the input to be entangled with an auxiliary system not taking part in the exchange. The trace distance between pure states does not exceed the corresponding vector distance; purifying a mixed input and then taking the partial trace gives the system error bound of Eq. (35).

Applying the same pointer readout to both outputs, write their unnormalized conditional states as $\mathcal{I}_r^{(L)}(\rho)$ and $\mathcal{I}_r^{\text{id}}(\rho)$. Neither the readout nor discarding the reference increases the trace distance, so the outputs carrying orthogonal outcome labels satisfy

$$\frac{1}{2} \sum_r \|\mathcal{I}_r^{(L)}(\rho) - \mathcal{I}_r^{\text{id}}(\rho)\|_1 \leq \delta_L.$$

Taking the trace of each term gives the total-variation bound of the complete outcome distribution. If $p_r^{\text{id}} > \delta_L$, then from $p_r^{(L)} \geq p_r^{\text{id}} - \delta_L$ and the triangle inequality for the normalization one also obtains

$$D(\rho_r^{(L)}, \rho_r^{\text{id}}) \leq \frac{2\delta_L}{p_r^{\text{id}} - \delta_L}.$$

A finite error therefore provides no uniform control of the normalized state for conditional outcomes of very low share.

When the same independent reference is prepared anew before each exchange, the difference of the iterated maps expands as

$$\mathcal{E}_L^n - \mathcal{E}_{\text{id}}^n = \sum_{j=0}^{n-1} \mathcal{E}_L^{n-1-j} (\mathcal{E}_L - \mathcal{E}_{\text{id}}) \mathcal{E}_{\text{id}}^j.$$

By the contractivity of the trace distance, the accumulated error does not exceed $n\delta_L$; the same holds when the same free propagation action is composed or the readout labels are kept. Under a non-flat dispersion, the initial reference state with $L > 1$ has energy coherence and is in general not a KMS state; its mean energy is $(L+1) \sum_j \Delta E_j / 2$. Improving the accuracy requires enlarging the preparation range; the estimates above do not support unlimited reuse of a finite reference, nor do they bring a nonthermal detection preparation into the same-temperature equilibrium conclusions. The joint exchange always balances the total energy; for position outcomes that do not resolve the exchanged energy, the readout of conservation uses the general mean balance of §5.2, and no separate definite Bohr frequency can be assigned to each pointer outcome.

Finally, consider one position-record action, expressing the scanned intensities as shares of the detection outcomes with the same incident normalization. For each phase $|I_L - I_{\text{id}}| \leq \delta_L$, so the deviations of the maximum and of the minimum each do not exceed δ_L . Let $u = I_{\text{max}}^{(L)} - I_{\text{max}}^{\text{id}}$, $w = I_{\text{min}}^{(L)} - I_{\text{min}}^{\text{id}}$, and $s_{\text{id}} = I_{\text{max}}^{\text{id}} + I_{\text{min}}^{\text{id}}$. Putting the difference over a common denominator gives

$$\mathcal{V}_{\text{fr}}^{(L)} - \mathcal{V}_{\text{fr}}^{\text{id}} = \frac{2(I_{\text{min}}^{\text{id}} u - I_{\text{max}}^{\text{id}} w)}{s_{\text{id}}(s_{\text{id}} + u + w)}.$$

For $s_{\text{id}} > 2\delta_L$ the absolute value of the numerator does not exceed $2s_{\text{id}}\delta_L$ and the denominator is at least $s_{\text{id}}(s_{\text{id}} - 2\delta_L)$, which gives the visibility error bound of the main text. The non-selective action,

the environment overlap, and the dichotomous-record relation in the target model are still those computed in §6.1; the finite conserving realization is compared with them through these bounds.

Shell transport at the same time involves the change of position space on adjacent shells. The relation between the shell shift and momentum can be computed directly on the interior shells. The number of principal-unit positions on shell j is $N_j = p^{K-j-1}$. Fixing the species label and writing the integer Fourier index as n , define

$$e_n^{(j)} = N_j^{-1/2} \sum_{\ell=0}^{N_j-1} e^{2\pi i n \ell / N_j} |j, \ell\rangle.$$

For $0 \leq j \leq K-2$, take

$$V|j, \ell\rangle = p^{-1/2} |j+1, \ell \bmod N_{j+1}\rangle.$$

Substituting $\ell = \ell' + tN_{j+1}$ with $t = 0, \dots, p-1$ gives

$$V e_n^{(j)} = p^{-1/2} N_j^{-1/2} \sum_{\ell'}^{p-1} \sum_{t=0}^{p-1} e^{2\pi i n t / p} e^{2\pi i n \ell' / N_j} |j+1, \ell'\rangle = [p | n] e_{n/p}^{(j+1)}.$$

The inner sum is $p[p | n]$, which gives the corresponding normalization. Hence $V^\dagger e_{n'}^{(j+1)} = e_{pn'}^{(j)}$. Define the dimensionless phase and the physical momentum

$$\theta_{j,n} = \frac{2\pi n}{N_j}, \quad k_{j,n} = v\theta_{j,n}.$$

Every nonzero shift matrix element satisfies $\theta_{j,n} = \theta_{j+1, n/p}$, and the physical momentum is preserved as well. If the unified energy calibration of reference [4] is adopted and on this restriction one specifies

$$H = \sum_{j,s,n} [E_s(k_{j,n}) + j \log p] |j, s, n\rangle \langle j, s, n|,$$

then the energy difference of every nonzero matrix element is the shell quantum $\log p$ expressed in that calibration, so $[H, V] = (\log p)V$ holds on the interior shells above. Here E_s takes the physical momentum as its argument, and this argument cannot be replaced directly by the dimensionless phase $\theta_{j,n}$. The terminal coupling must still be balanced separately, and this specification is not sufficient to make the coarse zero vector of Appendix A.1 an energy eigenstate.

Write the integer momentum-index transfer in a position record as d , with the corresponding modular momentum transfer $\Delta_{j,d} = 2\pi v d / N_j$ modulo $2\pi v$. Let $[n-d]_{N_j}$ denote the least nonnegative residue of $n-d$. Only the momentum-transfer unitary factor of the record action is taken here; the amplitude of the complete record is given separately by the matrix elements. With the sign convention that the environment gains the momentum,

$$J_d^{(j)} = e^{-2\pi i d \ell / N_j} = e^{-i \Delta_{j,d} x_\ell}, \quad J_d^{(j)} e_n^{(j)} = e_{[n-d]_{N_j}}^{(j)}.$$

The actual momentum difference of this finite-ring action is

$$k_{j,n} - k_{j,[n-d]_{N_j}} = \Delta_{j,d} + 2\pi v \left\lfloor \frac{n-d}{N_j} \right\rfloor.$$

The balance of the ordinary additive total momentum uses this actual difference. If the same environment excitation is required to carry a fixed $\Delta_{j,d}$, the participating modes must be restricted so that no index wraps around; transfers including wrap-around must have the corresponding difference

taken over by the environment. The intertwining relations below are for the modular index action on the finite ring. If $p \mid d$,

$$VJ_d^{(j)} = J_{d/p}^{(j+1)}V.$$

If $p \nmid d$, one can take $p \nmid n$ with $p \mid (n - d)$, so that $Ve_n = 0$ while $VJ_de_n \neq 0$, and the intertwining relation fails. Shell transport and position transport therefore cannot in general be interchanged.

Each record operator of the complete cycle is always obtained from $J_{rj} = \sqrt{\pi_j} \langle r|U|j \rangle$. The contributions of different complete environment final states are added at the level of shares; different paths landing on the same final state must have their amplitudes added before the modulus is squared. If the first-order amplitudes of weak couplings land on orthogonal single-record final states, their second-order shares can be counted separately; higher-order composite records and interference must be computed from the actual $YF^\dagger TF$ and cannot be inferred from the product of the probabilities of three independent processes.

A.3 Relaxation of Occupation

The cycle-by-cycle action can be written as a finite transfer chain only when the occupation of the system evolves autonomously. Take a common system basis $\{|a\rangle\}$ and write all Kraus operators of the actual cycle as $\{J_\ell\}$, the index ℓ including the incident and outgoing labels. Writing their matrix elements as $J_{\ell,ab}$,

$$[\mathcal{E}(\rho)]_{aa'} = \sum_{\ell,b,c} J_{\ell,ab} \rho_{bc} \overline{J_{\ell,a'c}}.$$

A diagonal initial state remains diagonal under this action if and only if

$$\sum_{\ell} J_{\ell,ab} \overline{J_{\ell,a'b}} = 0 \quad (a \neq a', \text{ every } b). \quad (48)$$

Necessity follows by substituting each $|b\rangle\langle b|$ in turn; sufficiency by the linear expansion of diagonal states. In this case define

$$P_{ab} = \sum_{\ell} |J_{\ell,ab}|^2.$$

From $\sum_{\ell} J_{\ell}^\dagger J_{\ell} = \mathbf{1}$ one has $P_{ab} \geq 0$ and $\sum_a P_{ab} = 1$, so the occupation of a diagonal initial state is iterated by the column-stochastic matrix P .

If the occupation of an arbitrary initial state is required not to depend on the coherence components, one must in addition have

$$\sum_{\ell} J_{\ell,ab} \overline{J_{\ell,ac}} = 0 \quad (b \neq c, \text{ every } a). \quad (49)$$

This is the condition that all off-diagonal input coefficients in $[\mathcal{E}(\rho)]_{aa}$ vanish; its necessity is verified by varying the real and imaginary parts of ρ_{bc} separately. This condition and Eq. (48) prescribe, respectively, the autonomous evolution of the occupation and the preservation of diagonal states.

When the same-temperature Gibbs state is diagonal in this basis, Eq. (9) makes its occupation a stationary distribution of P . If the finite chain is irreducible and aperiodic, this stationary distribution is unique and any initial occupation tends to it[3]. Convergence of the complete quantum state further requires that the remaining operator components contain no persistent eigenmode of modulus one; the convergence condition for occupation cannot replace this check.

The terminal occupation at zero temperature can be judged directly by an absorption condition. Suppose the occupation forms a closed finite chain, the subspace $\mathcal{D}$ spanned by the loss-free modes corresponds to the absorbing terminal, and every other state has a path of positive probability

reaching the terminal. Since the number of states is finite, one path can be chosen for each non-terminal state, with length at most L and with the product of probabilities along each path bounded below by the same constant $a > 0$. Once the terminal is reached it is never left, so any non-terminal initial state has at least a share a entering the terminal within L steps. Applying this bound segment by segment, the share still outside the terminal after nL steps is at most

$$(1 - a)^n \longrightarrow 0.$$

This proof gives the convergence of the terminal occupation and does not require the terminals to have equal final weights.

The coherence among the loss-free modes is still determined by the cycle phases. In the pure-incidence no-record decomposition of §2, if $J_r \psi_a = 0$ and $M \psi_a = e^{i\varphi_a} \psi_a$, then

$$\mathcal{E}^n(|\psi_a\rangle\langle\psi_b|) = e^{in(\varphi_a - \varphi_b)} |\psi_a\rangle\langle\psi_b|.$$

The coherence amplitude is preserved within $\mathcal{D}$, and different phases keep it oscillating. The entry of the zero-temperature occupation into the terminal and the convergence of the complete density operator to a steady state are therefore two conclusions to be judged separately.

A.4 Thermalization and Change of Identity

A particle changing its momentum in a medium can keep its existing identity; an admissible transit terminates the original identity, and the reverse exchange can replenish the occupation of that identity. Following the cycle of §2, let two momentum modes of the same identity take part in transport and transit. The chosen transit and transport matrix elements are inputs of this realization, and all shares and convergence conclusions are obtained from their complete composition.

Take two orthogonal eigenmodes $|a_0\rangle, |a_1\rangle$ of the same particle species on the same propagation branch, with momenta k_0, k_1 taken from the Fourier momentum set of the finite position lattice used, and energies $E_0 < E_1$. On the positive-frequency branch of §2,

$$E_i = v \arccos[\cos(m_\alpha/v) \cos(k_i/v)], \quad U_0|a_i\rangle = e^{i\varphi_i}|a_i\rangle, \quad \varphi_i = E_i/v.$$

Add the background state $|z\rangle$ and the final state $|b\rangle$ of another identity; the four states are orthogonal, with energies $E_z = 0 < E_b < E_0$. Write

$$P_i = |a_i\rangle\langle a_i|, \quad P = P_0 + P_1, \quad \bar{P} = \mathbf{1} - P.$$

P is the projection onto the identity sector tracked in this finite realization; the momentum modes outside it and the other channels do not take part in the coupling here. Under the same phase convention, the final state b continues to propagate as $U_0|b\rangle = e^{iE_b/v}|b\rangle$, and the action on the background state z is taken to be the identity. The computation below is first carried out in the common eigenmode representation; applying the Fourier transforms at both ends of Eq. (2) simultaneously gives the cycle in the same position representation.

The environment of each cycle consists of two mutually independent parts uncorrelated with the system, taking part in transit and in transport respectively. The transit environment B_m has five orthogonal states, with energies and incident weights

Environment state	$ \text{vac}\rangle$	$ 1_i\rangle$	$ \nu_i\rangle \ (i = 0, 1)$
Energy	0	E_i	$E_i - E_b$
Weight	π_{vac}	$\pi_{\text{vac}} e^{-\beta E_i}$	$\pi_{\text{vac}} e^{-\beta(E_i - E_b)}$

where $\beta > 0$, $\pi_{\text{vac}} = Z_m^{-1}$, and

$$Z_m = 1 + \sum_{i=0}^1 \left[e^{-\beta E_i} + e^{-\beta(E_i - E_b)} \right].$$

$|1_i\rangle$ carries all the energy and charge flowing out of a_i , and $|\nu_i\rangle$ carries the admissible emitted pair. If the conserved charges of a_i, b are q_a, q_b , the charges of the two kinds of environment states are q_a and $q_a - q_b$, respectively; if the final-state momentum is k_b , their momenta are k_i and $k_i - k_b$, respectively. These assignments must still satisfy the final-state thresholds and selection relations of the identified channel.

The transport environment B_t has two states $|g\rangle, |e\rangle$ with energies 0, ΔE , where $\Delta E = E_1 - E_0$. The same temperature gives

$$q_g = \frac{1}{1 + e^{-\beta \Delta E}}, \quad q_e = \frac{e^{-\beta \Delta E}}{1 + e^{-\beta \Delta E}}.$$

The excitation e carries the momentum difference $\Delta k = k_1 - k_0$ and keeps the charge balance required for preserving the particle identity; it is an excitation of the medium and is not required to have the dispersion of a free photon. The physical incident environment is $\omega_{\beta, m} \otimes \omega_{\beta, t}$, both parts being Gibbs states of their finite energy operators.

Transit uses two mutually orthogonal three-state blocks

$$e_{1i} = |a_i, \text{vac}\rangle, \quad e_{2i} = |z, 1_i\rangle, \quad e_{3i} = |b, \nu_i\rangle.$$

The total energy of the three states of each block is E_i . The two momentum modes take the resonance ratio of the same exchange station, $r_{a_i} = R_s = ce^{i\vartheta}$ with $0 < c < 1$ and $X = 1 - c^2$; this resonance ratio is still evaluated at the fixed amplitude-level evaluation point. The outflow F is defined as in §2, and the transit generator is taken as

$$K_{\text{mid}} = \eta \sum_{i=0}^1 (|e_{3i}\rangle\langle e_{2i}| + |e_{2i}\rangle\langle e_{3i}|), \quad 0 < \eta < \frac{\pi}{2}.$$

Let $W = F^\dagger e^{-iK_{\text{mid}}} F$, each factor being the identity on the unlisted orthogonal complement. Transport exchanges only the two momentum modes of the same identity,

$$Y = \exp[-i\xi(|a_0, e\rangle\langle a_1, g| + |a_1, g\rangle\langle a_0, e|)], \quad 0 < \xi < \frac{\pi}{2},$$

and is the identity on the remaining joint states. Extending W, Y to the other environment factor, the complete cycle is

$$U = (U_0 \otimes \mathbf{1})Y(W \otimes \mathbf{1}_{B_t}).$$

The dimensions of the system and of the two environment parts are 4, 5, and 2, respectively, and the joint dimension is 40. Every nonzero exchange matrix element connects joint states of the same total energy, and U_0 commutes with the system energy, so U is unitary and commutes with the total energy; the charge and momentum matching above is likewise preserved term by term. By Eq. (9), the complete map preserves

$$\sigma_\beta = Z_S^{-1} \text{diag}(1, e^{-\beta E_b}, e^{-\beta E_0}, e^{-\beta E_1}).$$

Writing $\pi_P = \text{Tr}(P\sigma_\beta)$, the conditional Gibbs state of this identity is

$$\sigma_P = \frac{P\sigma_\beta P}{\pi_P} = q_g P_0 + q_e P_1.$$

The complete Gibbs state is invariant; the equilibrium and convergence of the conditional state are determined by the non-exit sub-map.

A first-exit record in a positive-temperature environment requires the actual incidence to be resolved. Prepare the transit environment in the orthogonal basis above and keep its preparation label j_m ; after the cycle, read out the final-state label r_m . Keeping the incident label neither changes the non-selective Gibbs state of the physical environment nor feeds back on the system. For a history starting in P , $(j_m, r_m) = (\text{vac}, \nu_i)$ and $(\text{vac}, 1_i)$ mark the exits by the emitted pair and by the unretrieved single quantum respectively. The outgoing label alone is not sufficient for this distinction, because $1_i, \nu_i$ may also have been present in the positive-temperature incident environment from the start. All outcomes of transport are kept in the non-exit history. The proof of complete Gibbs invariance uses the initial state of the system and the physical environment; the recording device that keeps the preparation label is not additionally assumed to be in a joint Gibbs state.

For vacuum incidence, applying outflow, transit, and retrieval in turn gives

$$W e_{1i} = a_\eta e_{1i} - ic\sqrt{X}(1 - \cos \eta)e_{2i} + e^{i\vartheta}\sqrt{X} \sin \eta e_{3i}, \quad a_\eta = c^2 + X \cos \eta. \quad (50)$$

The environment final states of the three terms are orthogonal. The corresponding exit shares for vacuum incidence are

$$d_{\text{pair}} = X \sin^2 \eta, \quad d_{\text{single}} = c^2 X (1 - \cos \eta)^2, \quad d_{\text{pair}} + d_{\text{single}} = 1 - a_\eta^2.$$

The last identity follows by direct expansion from $c^2 + X = 1$. Under the actual thermal incidence,

$$\varepsilon_{\text{pair}} = \pi_{\text{vac}} d_{\text{pair}}, \quad \varepsilon_{\text{single}} = \pi_{\text{vac}} d_{\text{single}}, \quad \varepsilon = \varepsilon_{\text{pair}} + \varepsilon_{\text{single}}. \quad (51)$$

The joint states of a non-vacuum environment with any a_i all lie in the complement on which W acts as the identity; for vacuum incidence, the retrieval amplitudes of the two momentum modes are both a_η . Hence, for any operator ρ supported on P , including its off-diagonal components, the non-exit action of transit and retrieval is exactly

$$\mathcal{Q}_m(\rho) = [1 - \pi_{\text{vac}} + \pi_{\text{vac}} a_\eta^2] \rho = (1 - \varepsilon) \rho.$$

This scalar action is determined by the identical parameters of the two transit blocks.

Write the reduced map of the transport Y as $\mathcal{T}_\beta$, and let $c_\xi = \cos \xi$, $s_\xi = \sin \xi$. Its four Kraus operators on P are

$$\begin{aligned} K_{gg} &= \sqrt{q_g}(P_0 + c_\xi P_1), & K_{eg} &= -i\sqrt{q_g}s_\xi|a_0\rangle\langle a_1|, \\ K_{ee} &= \sqrt{q_e}(c_\xi P_0 + P_1), & K_{ge} &= -i\sqrt{q_e}s_\xi|a_1\rangle\langle a_0|. \end{aligned}$$

The first subscript is the outgoing and the second the incident label. From $q_g + q_e = 1$ and $c_\xi^2 + s_\xi^2 = 1$, the four terms satisfy $\sum K^\dagger K = P$. In the order of the complete cycle, the non-exit sub-map and its conditionally normalized action are

$$\mathcal{Q} = (1 - \varepsilon) \text{Ad}_{U_0} \circ \mathcal{T}_\beta, \quad \hat{\mathcal{Q}} = \frac{\mathcal{Q}}{1 - \varepsilon} = \text{Ad}_{U_0} \circ \mathcal{T}_\beta. \quad (52)$$

Here $\text{Ad}_{U_0}(\rho) = U_0 \rho U_0^\dagger$. This $\mathcal{Q}$ includes the identity-preserving thermal transport records. The conditional normalization only removes the common residual share after transit; the transport probabilities are still determined by $\mathcal{T}_\beta$.

Summing the four Kraus terms $K\rho K^\dagger$ gives the action on the complete density matrix:

$$\mathcal{T}_\beta \begin{pmatrix} x & w \\ \bar{w} & y \end{pmatrix} = \begin{pmatrix} (1 - q_e s_\xi^2)x + q_g s_\xi^2 y & c_\xi w \\ c_\xi \bar{w} & q_e s_\xi^2 x + (1 - q_g s_\xi^2)y \end{pmatrix}.$$

The ratio of the ascent and descent shares of the occupation is $q_e/q_g = e^{-\beta\Delta E}$. Substituting σ_P gives $\hat{\mathcal{Q}}(\sigma_P) = \sigma_P$. Furthermore, σ_P , $P_0 - P_1$, $|a_0\rangle\langle a_1|$, $|a_1\rangle\langle a_0|$ form a basis of the four-dimensional operator space, with eigenvalues respectively

$$1, \quad c_\xi^2, \quad c_\xi e^{i(\varphi_0 - \varphi_1)}, \quad c_\xi e^{-i(\varphi_0 - \varphi_1)}. \quad (53)$$

Hence, for any normalized initial state ρ_0 , the conditional state $\rho_n = \hat{\mathcal{Q}}^n(\rho_0)$ satisfies

$$(\rho_n)_{00} = q_g + c_\xi^{2n}[(\rho_0)_{00} - q_g], \quad (\rho_n)_{01} = [c_\xi e^{i(\varphi_0 - \varphi_1)}]^n (\rho_0)_{01}.$$

With $0 < c_\xi < 1$ both the occupation deviation and all coherence components decay, and the complete conditional state tends to the unique σ_P . The exact cycle rates of occupation relaxation and of coherence decay are, respectively,

$$\Gamma_{\text{pop}} = -2v \log c_\xi, \quad \Gamma_{\text{coh}} = -v \log c_\xi.$$

The slowest relaxation of a general initial state is decided by the latter; only an initial state without coherent deviation relaxes at the occupation rate alone.

The maintenance share of the identity is given by the same sub-map. Iterating Eq. (52),

$$\mathcal{Q}^n(\rho_0) = (1 - \varepsilon)^n \rho_n, \quad P_P(n) := \text{Tr } \mathcal{Q}^n(\rho_0) = (1 - \varepsilon)^n. \quad (54)$$

Let N_P be the cycle number of the first exit and $T_P = N_P/v$ the corresponding waiting time. With the two kinds of final states resolved,

$$\text{Pr}(N_P = n, r) = \varepsilon_r (1 - \varepsilon)^{n-1}, \quad r \in \{\text{pair}, \text{single}\}; \quad \mathbb{E}T_P = \frac{1}{v\varepsilon}, \quad \Gamma_{\text{out}} = -v \log(1 - \varepsilon).$$

Summing the first formula over $n \geq 1$ gives the probability $\varepsilon_r/\varepsilon$ that the first exit belongs to that kind; the two kinds sum to one. The mean waiting time and the exponential time scale follow the definitions of §4.2. The physical distinction between the two kinds of exit and the identification of the actual lifetime are treated as in §6.2; particles formed anew after an exit are counted separately.

Reverse formation is determined by the matrix elements of the same W . Computing the components of the other two columns on e_{1i} directly gives

$$\langle e_{1i}|W|e_{2i}\rangle = ic\sqrt{X}(1 - \cos \eta), \quad \langle e_{1i}|W|e_{3i}\rangle = -e^{-i\vartheta}\sqrt{X}\sin \eta.$$

Their squared moduli are d_{single} and d_{pair} , respectively. When system and environment are at the same temperature, the joint incident weights from $z \otimes 1_i$ and $b \otimes \nu_i$ are equal:

$$\frac{1}{Z_S} \pi_{\text{vac}} e^{-\beta E_i} = \frac{e^{-\beta E_b}}{Z_S} \pi_{\text{vac}} e^{-\beta(E_i - E_b)}.$$

The two kinds of incidence form an incoherent thermal mixture, so the formation share returning to a_i is

$$(A_m)_{ii} = \frac{\pi_{\text{vac}} e^{-\beta E_i}}{Z_S} (d_{\text{single}} + d_{\text{pair}}) = \varepsilon \frac{e^{-\beta E_i}}{Z_S}.$$

The complete formation input is therefore $A_m = \varepsilon \pi_P \sigma_P$. Transport and the single-cycle transfer both preserve σ_P , so the formation input after them still satisfies

$$A_P := P \mathcal{E}_\beta(\overline{P} \sigma_\beta \overline{P}) P = \varepsilon \pi_P \sigma_P. \quad (55)$$

The reverse term of the unretrieved single quantum is of fourth order for small η , but it must be kept for the exact balance.

This equality also shows the connection between complete equilibrium and conditional equilibrium. Since $[P, \sigma_\beta] = 0$ and the exit branches of this realization land in $\overline{P}$, the complete invariance projected onto P reads

$$\pi_P \sigma_P = \pi_P \mathcal{Q}(\sigma_P) + A_P.$$

Under a constant exit share ε per cycle, the conditional equilibrium $\hat{\mathcal{Q}}(\sigma_P) = \sigma_P$ is equivalent to $A_P = \varepsilon \pi_P \sigma_P$. The invariance of the complete Gibbs state does not by itself guarantee the latter equality; the reverse matrix elements above realize it here term by term. At complete equilibrium the total shares of formation and of exit per cycle are both $\varepsilon \pi_P$; the expectations of the event counts add over the number of cycles and give the equilibrium occupation relation of §6.3.

Whether thermalization can occur during identity maintenance depends on the time scales of conditional relaxation and of the first exit. Let $d_0 = (\rho_0)_{00} - q_g$ and $w_0 = (\rho_0)_{01}$. The difference between the conditional state and the equilibrium state is a traceless Hermitian 2×2 matrix whose two eigenvalues are negatives of each other, hence

$$\frac{1}{2} \|\rho_n - \sigma_P\|_1 = \sqrt{d_0^2 c_\xi^{4n} + |w_0|^2 c_\xi^{2n}} \leq c_\xi^n \frac{1}{2} \|\rho_0 - \sigma_P\|_1.$$

For a given error tolerance $0 < \delta_{\text{tol}} < 1$, a sufficient number of relaxation cycles is

$$n_{\text{mix}}(\delta_{\text{tol}}) = \max \left\{ 1, \left\lceil \frac{\log(1/\delta_{\text{tol}})}{-\log c_\xi} \right\rceil \right\}.$$

If $n_{\text{mix}}[-\log(1-\varepsilon)] \ll 1$, the maintenance share of the original identity is still close to one by the time the conditional state is close to equilibrium. For fixed ξ and small nonzero η , $\varepsilon = \pi_{\text{vac}} X \eta^2 + O(\eta^4)$, while n_{mix} does not change with η , so this separation of time scales can hold. For $\eta = 0$ there is neither exit nor reverse formation, while transport can still thermalize the conditional state; for $\xi = 0$ there is no occupation relaxation; for $\xi = \pi/2$ the conditional state becomes σ_P after one complete cycle, and the divergence of the negative logarithm does not mean a vanishing physical time.

This finite realization uses identical transit parameters for the two momentum modes and the environment couplings listed. Its identification with a specific particle process requires the admissible final states, the excitation spectrum of the medium, and the actual matrix elements to be compatible; the conditional convergence on the two modes does not by itself determine the convergence of the whole momentum distribution.

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